ECG-BASED DETECTION OF MYOCARDIAL INFARCTION AND ISCHEMIC ABNORMALITIES USING DEEP LEARNING AND WEB DEPLOYMENT

A project report submitted in partial fulfilment of the requirement for the Award of the Degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

by

Mohammed Abdullah (160722733056)

Mohammed Faisal Mohiuddin (160722733016)

Mohammed Hussain Moheet (160722733029)

Under the Guidance of
Ms. Sana Mateen
Assistant Professor, Dept. of CSE

Department of Computer Science and Engineering

Methodist College of Engineering and Technology,
King Koti, Abids, Hyderabad-500001.

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Department of Computer Science and Engineering

DECLARATION BY THE CANDIDATES

We **Mohammed Abdullah (160722733056), Mohammed Faisal Mohiuddin (160722733016)** and **Mohammed Hussain Moheet (160722733029),** students of Methodist College of Engineering and Technology, pursuing bachelor's degree in Computer Science and Engineering, hereby declare that this major Project report entitled **"ECG-BASED DETECTION OF MYOCARDIAL INFARCTION AND ISCHEMIC ABNORMALITIES USING DEEP LEARNING AND WEB DEPLOYMENT"** carried out under the guidance of **Ms. Sana Mateen** submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Computer Science and Engineering. This is a record work carried out by us and the results embodied in this report have not been reproduced/copied from any source.

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CERTIFICATE BY THE SUPERVISOR

This is to certify that this Major Project work entitled **"ECG-BASED DETECTION OF MYOCARDIAL INFARCTION AND ISCHEMIC ABNORMALITIES USING DEEP LEARNING AND WEB DEPLOYMENT"** ***by*** **Mohammed Abdullah (160722733056), Mohammed Faisal Mohiuddin (160722733016)** and **Mohammed Hussain Moheet (160722733029),** submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Computer Science and Engineering, during the academic year 2025-2026, is a Bonafide record of work carried out by them.

Ms. Sana Mateen,
Assistant Professor,
Dept. of CSE

Date:

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Department of Computer Science and Engineering

CERTIFICATE BY THE HEAD OF THE DEPARTMENT

This is to certify that this Major Project work entitled **"ECG-BASED DETECTION OF MYOCARDIAL INFARCTION AND ISCHEMIC ABNORMALITIES USING DEEP LEARNING AND WEB DEPLOYMENT"** ***by*** **Mohammed Abdullah (160722733056), Mohammed Faisal Mohiuddin (160722733016)** and **Mohammed Hussain Moheet (160722733029),** submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Computer Science and Engineering, during the academic year 2025-2026, is a Bonafide record of work carried out by them.

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Professor &
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Department of Computer Science and Engineering

PROJECT APPROVAL CERTIFICATE

This is to certify that this Major Project work entitled **"ECG-BASED DETECTION OF MYOCARDIAL INFARCTION AND ISCHEMIC ABNORMALITIES USING DEEP LEARNING AND WEB DEPLOYMENT"** ***by*** **Mohammed Abdullah (160722733056), Mohammed Faisal Mohiuddin (160722733016)** and **Mohammed Hussain Moheet (160722733029),** submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Computer Science and Engineering during the academic year 2025-2026, is a Bonafide record of work carried out by them.

INTERNAL **EXTERNAL** **HOD**

ACKNOWLEDGEMENTS

We would like to express our sincere gratitude to our project guide **Ms. Sana Mateen**, **Assistant Professor, CSE, Methodist College of Engineering and Technology,** for giving us the opportunity to work on this topic. It would never be possible for us to take this project to this level without her innovative ideas and her relentless support and encouragement.

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Our sincere thanks to **Dr. P. Lavanya, Professor** and **Head of the Department of Computer Science and Engineering,** for her valuable guidance and encouragement which has played a major role in the completion of the project and for helping us by being an example of high vision and pushing towards greater limits of achievement.

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We are indebted to the Department of Computer Science and Engineering and Methodist College of Engineering and Technology for providing us with all the required facility to carry our work in a congenial environment. We extend our gratitude to the CSE Department staff for providing us with the needful time to time whenever requested.

We would like to thank our parents for allowing us to realize our potential, all the support they have provided us over the years was the greatest gift anyone has ever given us and for teaching us the value of hard work and education. Our parents have offered us with tremendous support and encouragement, thanks to our parents for all the moral support and the amazing opportunities they have given us over the years.

Vision & Mission

VISION

To become a leader in providing Computer Science and Engineering education with emphasis on knowledge and innovation.

MISSION

M1: To offer flexible programs of study with collaborations to suit industry needs

M2: To provide quality education and training through novel pedagogical practices

M3: To Expedite high performance of excellence in teaching, research and innovations.

M4: To impart moral, ethical valued education with social responsibility.

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PEO2: Apply the knowledge of Computer Science Engineering to pursue higher education with due consideration to environment and society.

PEO3: Promote collaborative learning and spirit of teamwork through multidisciplinary projects

PEO4: Engage in life-long learning and developing entrepreneurial skills.

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At the end of 4 years, Computer Science and Engineering graduates at MCET will be able to:

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PSO2: Develop software applications with open-ended programming environments.

PSO3: Design and develop solutions by following standard software engineering principles and implement by using suitable programming languages and platforms

Program Outcomes (POs)

At the end of the four-year Bachelor of Engineering Program, engineering graduates will be able to:

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PO2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and the cultural, societal, and environmental considerations.

PO4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6. The engineer and society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7. Environment and sustainability: Understand the impact of professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8. Ethics: Apply ethical principles and commitment to professional ethics and responsibilities and norms of engineering practice.

PO9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, making effective presentations, and give and receive clear instructions.

PO11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

ABSTRACT

Heart diseases are one of the leading causes of death around the world, and myocardial infarction (MI), commonly known as a heart attack, is one of its most serious forms requiring rapid detection. Doctors use electrocardiogram (ECG) signals to identify signs such as ST-segment changes, T-wave inversions, and abnormal Q waves, but accurate interpretation demands significant clinical expertise that is not always available in every healthcare setting. This paper presents an end-to-end deep learning system for automated detection of myocardial infarction and ischemic abnormalities from ECG signals, incorporating explainability and web deployment. Three one-dimensional convolutional neural network (CNN) models were trained on the PTB-XL dataset (21,799 records, 500 Hz, 10 seconds): a 12-Lead CNN for full digital ECG, a Lead II CNN for single-lead ECG, and an Image CNN for scanned or photographed ECG paper recordings, with all models trained using the Adam optimizer over 50 epochs with a batch size of 64. The 12-Lead CNN achieved the best performance with a ROC-AUC of 0.8991, sensitivity of 0.8674, specificity of 0.7580, and F1 score of 0.8436, while the Lead II CNN achieved a ROC-AUC of 0.8528 and the Image CNN, trained on synthetic ECG images generated from PTB-XL signals, achieved a ROC-AUC of 0.8083. Explainability was integrated through Grad-CAM and saliency maps, allowing clinicians to visualize which parts of the ECG signal influenced each prediction. The complete system was deployed as a publicly accessible single-tier Streamlit Cloud web application supporting WFDB, CSV, and image inputs with automatic model routing, a clinical risk indicator, explainability visualizations, and a downloadable PDF report.

The proposed system demonstrates an effective, accessible, and scalable solution for AI-assisted cardiac diagnosis, aligning with Sustainable Development Goal (SDG) indicators 3.4.1, 3.8.1, 3.d.1, 9.5.1, and 9.c.1, contributing to improved healthcare outcomes and technological innovation.

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1. INTRODUCTION

Heart diseases are one of the leading causes of death worldwide. Myocardial infarction happens when blood supply to part of the heart is blocked, which causes damage to the heart muscle. If it is not detected quickly, the outcome for the patient gets much worse. The electrocardiogram is the main tool doctors use to check for MI. It records the electrical activity of the heart and shows changes in the waveform like ST-segment elevation, T-wave inversions, and abnormal Q waves that point to ischemia.

Reading an ECG properly requires a lot of training and experience. In many places, especially smaller hospitals and rural clinics, there are not enough trained cardiologists available all the time. This delay in interpretation can be very harmful for the patient.

Deep learning models, especially CNNs, have shown that they can classify ECG signals with good accuracy. But most of these systems only exist as research models. They have no interface that a real user can work with, and they do not explain why they made a prediction. Doctors need to understand the reasoning before they trust any system like this.

This project addresses all of these problems together. We built three CNN models to handle different types of ECG input, added Grad-CAM and saliency map explanations so clinicians can understand the predictions, and deployed the whole system as a working web application on Streamlit Cloud. The system accepts WFDB files, CSV files, and ECG images, automatically picks the right model for each input, and produces a structured PDF report with embedded visualizations.

The system was initially designed with a separate frontend and backend, where a Streamlit interface communicated with a FastAPI server for model inference. For deployment, this was simplified into a single-level architecture where Streamlit directly handles inference, reducing complexity and improving maintainability.

Users upload ECG inputs through the interface, which are processed using different approaches such as Lead II analysis, full 12 lead analysis, and image-based extraction. The system also applies explainability methods like Grad-CAM and saliency maps to highlight important signal regions and generates a simple report with the results.

The application is deployed on Streamlit Cloud using CPU, with an average inference time of 15 to 25 seconds. The project is modular, with separate components for the interface, inference logic, and trained models, making it easy to manage and extend.

1.1 Objective

The main goals of this project are listed below.

- To develop three CNN models: a Lead II CNN, a 12-Lead CNN, and an Image CNN for MI and ischemia detection.
- To train all models on the PTB-XL dataset with z-score normalization and a 70/15/15 patient-level train-validation-test split.
- To evaluate performance using ROC-AUC, sensitivity, specificity, F1 score, and accuracy.
- To integrate Grad-CAM and saliency maps so clinicians can see which parts of the ECG signal influenced the prediction.
- To generate synthetic ECG paper images from PTB-XL signals to train the Image CNN, since no public labelled ECG image dataset exists.
- To deploy the system as a single-tier Streamlit Cloud application with automatic input routing, clinical risk indicator, PDF report.
- To conduct an ablation study to verify key design decisions around sampling frequency, dataset size, and compute environment.

1.2 Scope

This project covers the full pipeline from raw ECG data to deployed web output. It includes data loading, preprocessing, model training, evaluation, explainability generation, deployment, and report generation.

Three input types are supported. WFDB files and 12-column CSV files go to the 12-Lead CNN. Single-column CSV files go to the Lead II CNN. Image and PDF files go through an image extraction pipeline and then to the Image CNN. The image pathway is marked as experimental because the model was trained on synthetic images and there is a known domain gap with real scanned ECGs.

The system is not a certified medical device. It is meant as a clinical decision support and screening tool. Real-time ECG monitoring, multi-class arrhythmia classification beyond MI and ischemia, and integration with hospital systems are not part of this project but are natural next steps.

2. LITERATURE SURVEY

A comprehensive review of existing research was conducted, focusing on deep learning techniques for ECG analysis, particularly myocardial infarction (MI) detection, ischemia classification, explainability in cardiac AI, and ECG image processing. A total of 30 research papers were analyzed, covering a wide range of methodologies, datasets, and practical implementations. The findings, along with their relevance to this work and identified research gaps, are summarized in Table 2.1.

Table 2.1: Literature Survey

No	Year	Paper Title and Authors	Ref	Summary	Relevance	Gap
1	2021	Acute Myocardial Infarction Detection Using Deep Learning, Chen et al.	[1]	ResNet CNN on PTB-XL. AUC about 0.96. Tested on external data.	Showed CNNs work well on PTB-XL.	Only MI, not ischemia. No deployment or explainability.
2	2022	Reliable Detection of Myocardial Ischemia Using Machine Learning, Zhao et al.	[2]	ECG and VCG features with SVM. Accuracy about 90%.	Showed classical ML can detect ischemia.	Needs VCG signals. Manual features required.
3	2022	Deep Learning Networks Accurately Detect ST Elevation, Wu et al.	[3]	Deep CNN predicts STEMI and culprit vessel. AUC above 0.97.	Showed DL can detect ischemia patterns.	Proprietary dataset. Cannot be reproduced.
4	2023	Hybrid CNN BiLSTM Model for MI Detection, Hasbullah et al.	[4]	CNN and BiLSTM on PTB-XL. Accuracy about 91%.	Showed hybrid models work for ECG time series.	No explainability and no deployment.
5	2024	Time Frequency Transform with Lightweight Model, Sheth et al.	[5]	ECGs to time-frequency domain then CNN-LSTM. Accuracy about 83%.	Inspired lightweight models for web deployment.	Moderate accuracy and limited dataset.
6	2024	Real Time MI Detection Using	[6]	CNN on edge device with	Showed deployable AI	Hardware specific. Not a

No	Year	Paper Title and Authors	Ref	Summary	Relevance	Gap
		Edge AI, Gragnaniello et al.		near real-time classification.	for ECG is possible.	general web solution.
7	2025	Multi Domain Feature Fusion CNN for MI, Chen et al.	[7]	Time, frequency, and time-frequency features. Accuracy about 99.9%.	Showed multi-domain CNNs can localize MI.	Very complex. No web demo available.
8	2022	Noise Robustness in MI Detection, Sraitih et al.	[8]	SVM, KNN, and Random Forest on PTB under noise. Random Forest most stable.	Highlighted importance of preprocessing.	No deep learning and no deployment.
9	2023	Inferior MI Detection from Lead II Using 2D CNN, Yousuf et al.	[9]	1D ECG to 2D image using GAF. Accuracy about 99.6% on Lead II.	Showed single lead detection is feasible.	Image conversion does not generalize to real ECG images.
10	2023	DL ECG Prediction of MI in Emergency Patients, Gustafsson et al.	[10]	Ensemble ResNets on 492k ECGs from Sweden. C-statistic 0.99 for STEMI.	Real-world clinical validation at scale.	Proprietary dataset. Not reproducible.
11	2020	Attentive RNN to Fuse 12-Lead ECG and Clinical Features, Prabhakararao and Dandapat	[11]	Attention-based BiLSTM with ECG and patient metadata. Accuracy about 98%.	Inspired use of attention for explainability.	Requires patient metadata alongside signal.
12	2024	Interpretable ECG Analysis for MI Using Counterfactuals, Pawelczyk et al.	[12]	Counterfactual explanations on PTB-XL.	Showed model transparency matters for clinical use.	Explanation only. No full detection pipeline.
13	2024	Deep Learning Approaches for MI: A Review, Radwa et al.	[13]	Reviewed CNN, RNN, and Transformer	Supported CNN as the base architecture choice.	Review only. No implementation.

No	Year	Paper Title and Authors	Ref	Summary	Relevance	Gap
				methods for MI detection.		
14	2020	PTB-XL: A Large Publicly Available ECG Dataset, Wagner et al.	[14]	Described the PTB-XL dataset, labeling, and standard splits.	This is the exact dataset we used.	Dataset paper only. Not a detection model.
15	2022	Automated Detection of MI and Conduction Disorders, Hammad et al.	[15]	Deep CNN on PTB-XL. AUC about 0.95 across multiple cardiac classes.	Showed multi-class ECG detection on PTB-XL is feasible.	No explainability and no web deployment.
16	2017	Grad-CAM: Visual Explanations from Deep Networks, Selvaraju et al.	[16]	Introduced Grad-CAM to highlight important CNN regions using gradients.	Used for explainability to visualize ECG regions influencing predictions.	Not specific to ECG. Does not improve model accuracy.
17	2023	ECG-Image-Kit: Synthetic ECG Image Generation Toolkit, Kshama et al.	[17]	Toolkit for generating synthetic ECG images for digitization research.	Informed our synthetic image generation pipeline.	Synthetic data may not match real-world ECG variability.
18	2019	Cardiologist-Level Arrhythmia Detection Using DNN, Hannun et al.	[18]	DNN classifying 12 rhythm classes from 91,232 ECGs. F1 0.837 exceeding cardiologist average.	Demonstrated end-to-end DL can achieve cardiologist-level ECG performance.	Arrhythmia rhythm classes only, not MI or ischemia.
19	2019	Inter- and Intra-Patient ECG Heartbeat Classification, Mousavi and Afghah	[19]	Sequence-to-sequence DL with CNN for heartbeat classification on MIT-BIH.	Demonstrated strong ECG classification with deep learning approaches.	Heartbeat-level arrhythmia only. Not MI detection.
20	2017	Deep CNN Model to Classify	[20]	9-layer deep CNN classifying 5	Demonstrated CNN effectiveness for	Heartbeat classification only. Not

No	Year	Paper Title and Authors	Ref	Summary	Relevance	Gap
		Heartbeats, Acharya et al.		ECG heartbeat categories. Accuracy 94.03% on MIT-BIH.	automated ECG heartbeat analysis.	applied to MI detection.
21	2018	Deep Learning for Healthcare Applications: A Review, Faust et al.	[21]	Review of 53 DL papers on ECG, EEG, EMG, and EOG signals (2008-2017).	Supported CNN as dominant architecture for physiological signal classification.	Review only. No original model or implementation.
22	2021	1D Convolutional Neural Networks and Applications: A Survey, Kiranyaz et al.	[22]	Survey of 1D CNN architectures for biomedical signals and other engineering domains.	Justified use of 1D CNN architecture for raw ECG waveform classification.	Survey only. Does not address MI detection or explainability.
23	2015	Adam: A Method for Stochastic Optimization, Kingma and Ba	[23]	Introduced the Adam optimizer for adaptive gradient-based optimization.	Adam optimizer used in training all three CNN models in our system.	General optimization algorithm. Not specific to ECG or medical AI.
24	2016	Learning Deep Features for Discriminative Localization, Zhou et al.	[24]	Introduced Class Activation Mapping (CAM) using global average pooling.	Foundational work for Grad-CAM used for ECG explainability in our system.	Designed for 2D images. Not specific to 1D ECG signals.
25	2000	PhysioBank, PhysioToolkit, and PhysioNet, Goldberger et al.	[25]	Describes PhysioNet: open-access repository hosting ECG databases including PTB-XL.	PTB-XL dataset used in our work is accessed via PhysioNet.	Infrastructure paper only. Not a detection model.
26	2001	The Impact of the MIT-BIH Arrhythmia	[26]	Described the MIT-BIH database and its	MIT-BIH database referenced in	Arrhythmia database only. Does not

No	Year	Paper Title and Authors	Ref	Summary	Relevance	Gap
		Database, Moody and Mark		role in standardizing ECG algorithm evaluation.	related work on ECG heartbeat classification.	address MI or ischemia.
27	2011	Scikit-learn: Machine Learning in Python, Pedregosa et al.	[27]	Open-source Python ML library providing classification and evaluation metrics tools.	Used to compute AUC, sensitivity, specificity, and F1 in our evaluation.	General ML library. Not specific to ECG or medical signals.
28	2021	Cardiovascular Diseases (CVDs): Key Facts, World Health Organization	[28]	WHO report citing 17.9 million annual CVD deaths globally.	Provides epidemiological justification for automated MI detection research.	Statistical report only. Not a technical contribution.
29	2019	PyTorch: High-Performance Deep Learning Library, Paszke et al.	[29]	Describes PyTorch framework for flexible deep learning model development.	All three CNN models were implemented and trained using PyTorch.	General DL framework. Not specific to ECG analysis.
30	2019	HuggingFace Transformers: State-of-the-Art NLP, Wolf et al.	[30]	Describes Hugging Face Hub for open-source model hosting and sharing.	Trained CNN model weights are hosted on Hugging Face Hub for deployment.	Model hosting platform. Not ECG-specific.

2.1 Gap Analysis

From the review of existing literature, several key limitations were consistently observed across the studied works.

Most approaches prioritize model accuracy with little attention given to interpretability, which is a critical requirement for clinical acceptance. While techniques such as Grad-CAM have been proposed in the broader deep learning community, they are rarely integrated into complete, end-to-end ECG analysis systems. Additionally, the majority of existing models remain at the research or experimental stage, with no accessible deployment or user interface, making them impractical for real-world clinical use.

Another significant limitation is the lack of input flexibility. Most systems are designed to handle only a single input format either multi-lead digital signals or single-lead recordings with no unified framework capable of processing multiple input types. Similarly, very few works address both myocardial infarction and ischemic abnormalities together within the same pipeline, as most studies focus exclusively on MI detection. The reliance on proprietary or institutional datasets in many high-performing studies further limits the reproducibility and fair comparison of results. Tools such as ECG-Image-Kit, while useful, focus solely on synthetic image generation without offering any detection or deployment component.

Our project fills these gaps by building three models in one pipeline, integrating Grad-CAM for explainability, deploying a working Streamlit Cloud application, and training everything on the publicly available PTB-XL dataset.

3. SYSTEM ANALYSIS

3.1 Existing System

Most existing automated ECG analysis systems fall into three categories. The first uses traditional machine learning methods like SVM, Random Forest, and KNN on hand-crafted ECG features. These are interpretable but require domain knowledge and do not generalize well. The second uses deep learning models that learn from raw ECG signals but are only available as research code with no deployment or explanation. The third converts 1D ECG signals into 2D image representations for classification. These show high accuracy in controlled settings but fail when the input is an actual scanned ECG image because of the domain mismatch.

3.1.1 Drawbacks of Existing Systems

- No deployment interface. Most models exist only as research code with no user interface.
- No explainability. Predictions are made without showing which ECG regions drove the result.
- Single input format. Existing systems support only one type of input.
- Limited to MI only. Very few systems handle both MI and ischemia together.
- No structured reports. Outputs are raw probability scores with no clinical documentation.
- Proprietary datasets. Many high-performing systems use hospital data that cannot be reproduced.

3.2 Proposed System

Our system is a three model CNN pipeline deployed as a Streamlit Cloud web application. It uses a single-level architecture where all inference functions are called directly from the frontend. This removes the need for a separate backend and makes the system easier to deploy. The system accepts three types of input and routes each one to the appropriate model. After prediction, Grad-CAM and saliency maps are generated to show which parts of the signal were important. The output is internally structured in JSON format for better organization and future integration. However, this JSON is not shown to the user. Instead, the results are displayed in a simple and clear way through the interface. A structured PDF report is also generated for every prediction.

3.2.1 Advantages of Proposed System

- Three input formats supported with automatic routing to the right model.
- Grad-CAM and saliency map explanations generated for every prediction.
- Single-tier architecture compatible with Streamlit Cloud free deployment.
- Publicly accessible at https://github.com/Abdullah1607/ECG-MI-Detection.
- Generates a downloadable PDF report output.
- Clinical risk indicator and separate patient and clinician interpretation tabs.
- Trained entirely on the publicly available PTB-XL dataset.

3.3 Applications

This system can be used in clinics, emergency departments, and telemedicine platforms where a cardiologist may not always be available for immediate ECG reading. It can also be used in medical colleges as a teaching tool where students can upload ECG recordings and study the model's predictions and explanations together.

3.4 Technical Specification

3.4.1 Hardware Requirements

Table 3.1: Hardware Requirements

Component	Specification
Processor	Intel Core i5 8th Gen or AMD Ryzen 5 or above
RAM	Minimum 8 GB (16 GB recommended for training)
GPU	NVIDIA CUDA GPU for training. CPU sufficient for inference on Streamlit Cloud.
Storage	Minimum 20 GB free space for dataset and model weight files
Operating System	Windows 10 or 11, Ubuntu 20.04 or above, macOS 11 or above
Network	Required for Streamlit Cloud deployment and GitHub access

3.4.2 Software Requirements

Table 3.2: Software Requirements

Component	Version or Details
Python	3.10 or above
PyTorch	2.0 or above (CPU for cloud, CUDA for local training)
Streamlit	1.25 or above
WFDB	4.1 or above for reading ECG files
OpenCV and Pillow	Image processing for the ECG image extraction pipeline
NumPy and Pandas	Latest stable versions
Matplotlib and Seaborn	Visualization of ECG signals and results
ReportLab	PDF report generation
Poppler	PDF to image conversion for the image pipeline
GitHub	Source code repository and Streamlit Cloud deployment trigger

4. SYSTEM DESIGN

4.1 System Architecture

The system uses a single-tier architecture where everything runs inside one Streamlit application. There is no separate backend server. When the user uploads a file, the Streamlit app handles routing, preprocessing, inference, explainability generation, and report creation all in one place. This design was chosen because Streamlit Cloud requires a single-application setup and does not support running a separate server process alongside it.

The pipeline flows like this, the user uploads a file through the Streamlit UI, then ECG Loader reads and parses the file, then input router selects the right model, then the CNN model runs inference, the Grad-CAM and saliency modules generate explanations, natural language interpretation text is created, a PDF report is assembled, and everything is shown in the Streamlit interface.

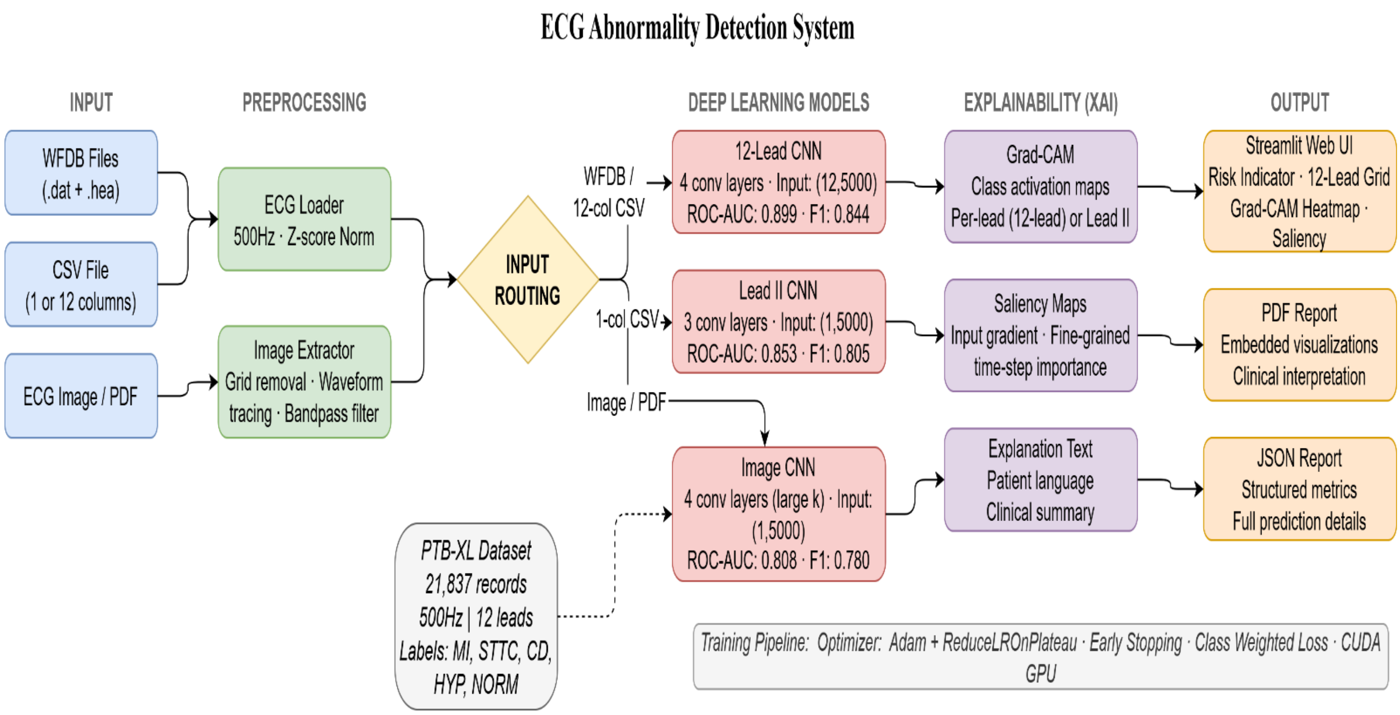


Fig. 4.1 – System Architecture

4.2 CNN Architectures

All three models use 1D CNN architectures with a fixed input of 5000-time steps. Each convolutional block has a 1D convolution, batch normalization, ReLU activation, max pooling with kernel size 2, and dropout at rate 0.3. The classification head has two fully connected layers with softmax output for binary classification (Normal vs Abnormal).

4.2.1 Lead II CNN

The Lead II CNN accepts input of shape (1, 5000). It has three convolutional blocks with filter sizes 32, 64, and 128 and kernel sizes 7, 5, and 5. After the three blocks the features are flattened and passed through two fully connected layers of size 256 and 1.

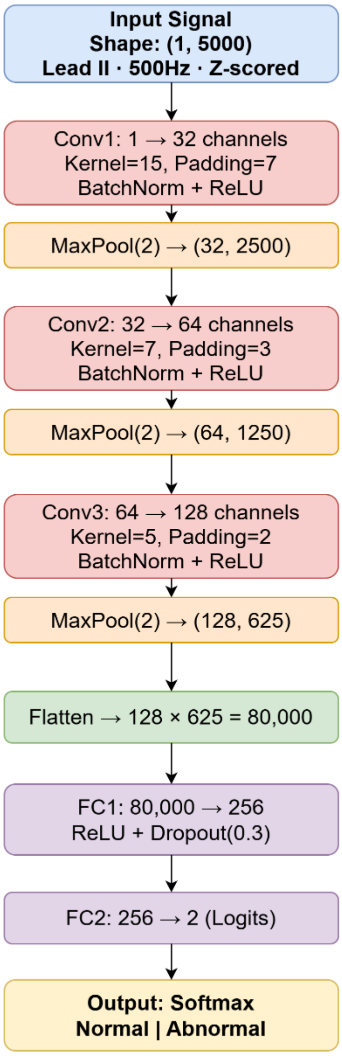


Fig. 4.2 – Lead II CNN Architecture

4.2.2 12-Lead CNN

The 12-Lead CNN accepts input of shape (12, 5000). It has four convolutional blocks with filter sizes 32, 64, 128, and 128 and kernel sizes 7, 5, 5, and 3. The extra block helps the model learn more complex patterns from the richer multi-lead data. The classification head is the same as the Lead II CNN.

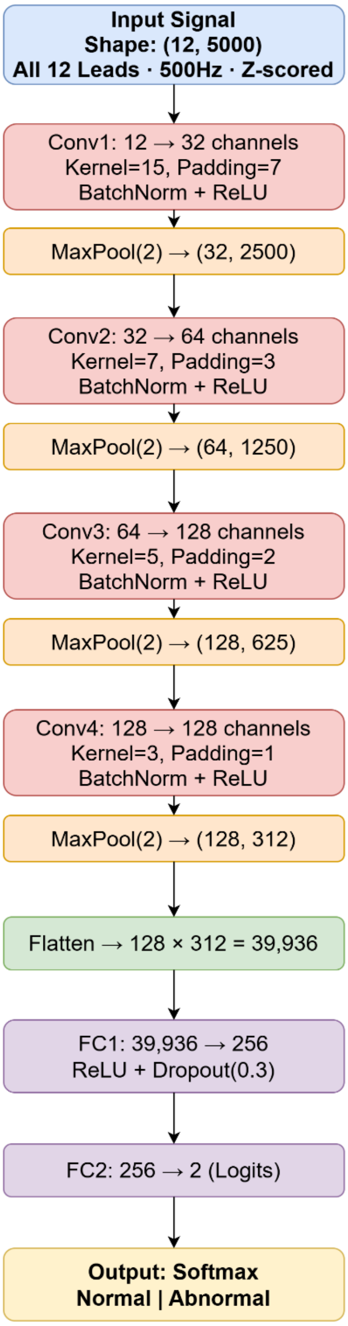


Fig. 4.3 – 12- Lead CNN Architecture

4.2.3 Image CNN

The Image CNN follows a similar 1D CNN architecture to the Lead II model, with the addition of an extra convolutional layer to better capture variations and noise introduced during the image extraction process.and accepts input of shape (1, 5000). The input comes from the image extraction pipeline rather than a digital ECG file. It was trained separately on synthetic ECG images generated from PTB-XL signals.

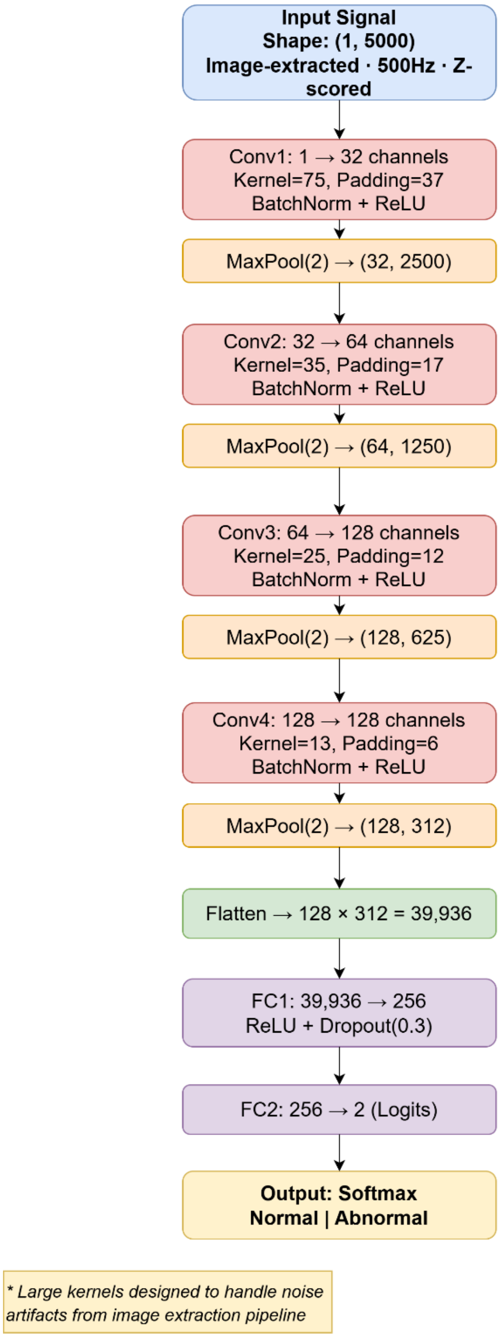


Fig. 4.4 – Image CNN Architecture

4.3 Input Routing Logic

The routing logic runs inside the Streamlit app. When a file is uploaded, the app checks its extension and structure and picks the right model. Table 4.1 shows the full routing table.

Table 4.1: Input Routing Table

Input Type	Model Used	AUC	Status
WFDB .dat and .hea files	12-Lead CNN	0.899	Primary
CSV with 12 columns	12-Lead CNN	0.899	Primary
CSV with 1 column	Lead II CNN	0.853	Primary
Image or PDF	Image CNN	0.808	Experimental

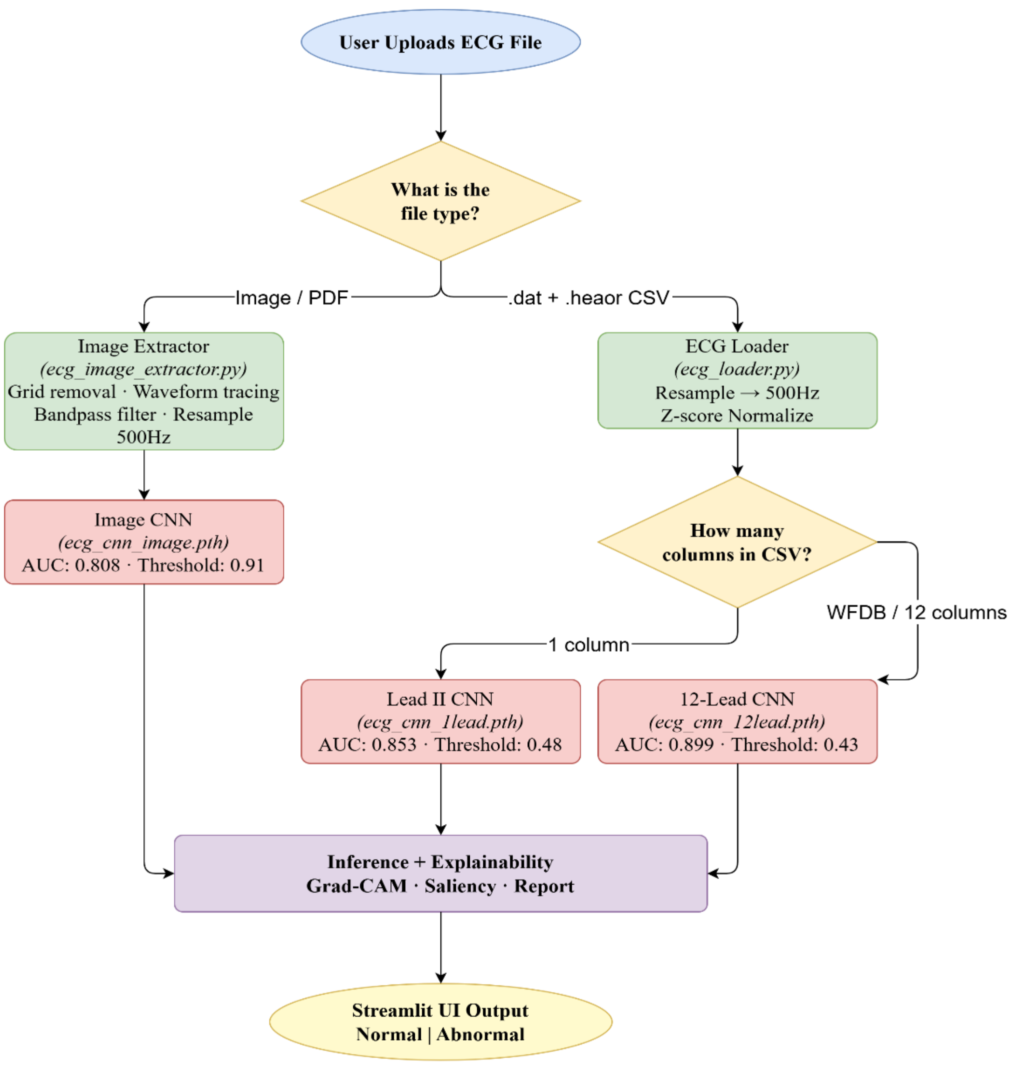


Fig. 4.5 – Input Routing Decision Logic.

4.4 Image Extraction Pipeline

When the user uploads an ECG image or PDF, the system runs a computer vision pipeline to extract the waveform before passing it to the Image CNN. The steps are: load the image and convert PDF files using Poppler; auto-rotate portrait images to landscape; convert to grayscale and denoise; apply adaptive thresholding; remove horizontal grid lines using morphological operations; detect the rhythm strip in the bottom 55 to 76 percent of the image; extract the waveform using a darkness-weighted centroid method; apply a bandpass filter from 0.5 to 40 Hz; resample to 5000 samples; and z-score normalize.

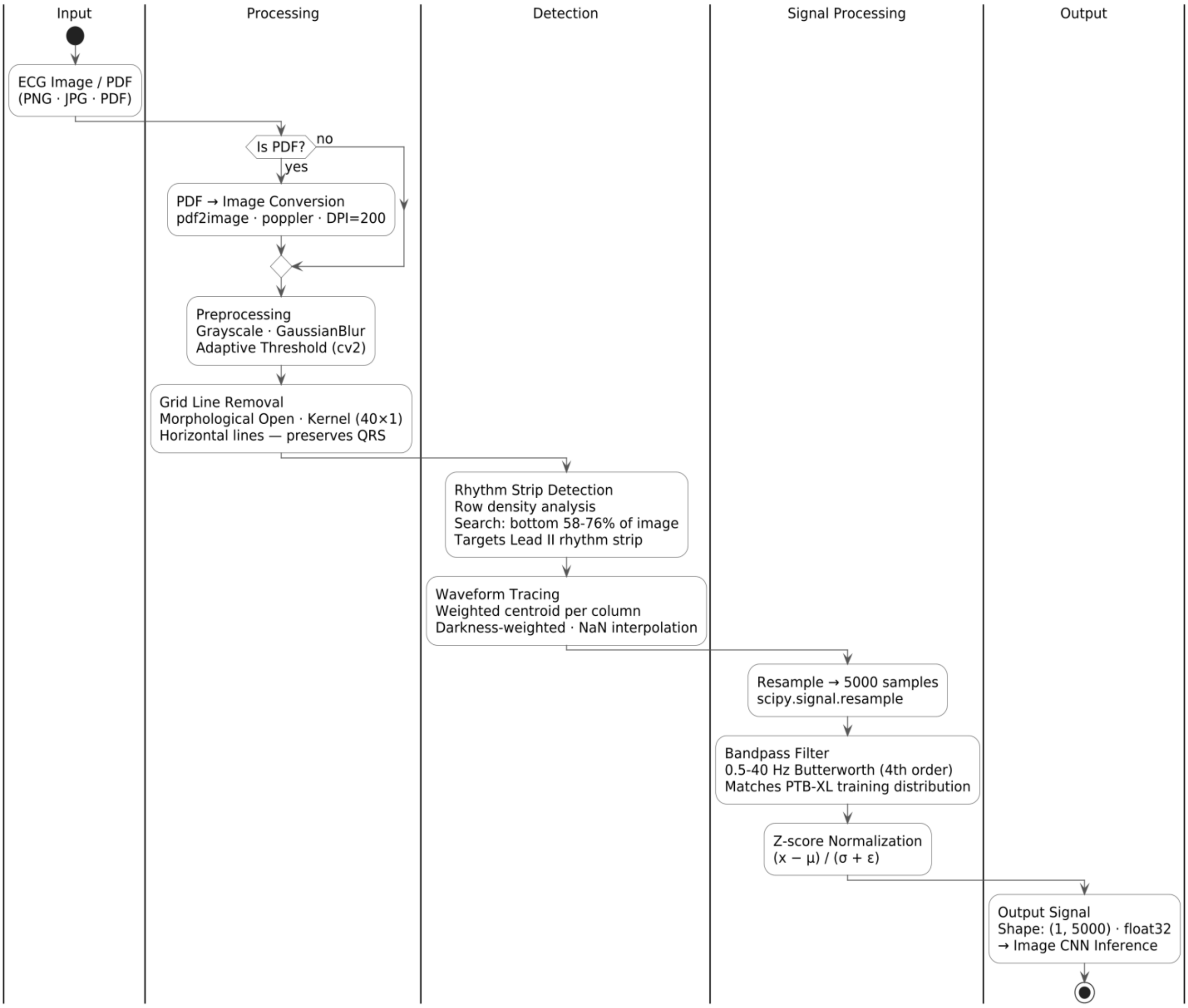


Fig. 4.6 - Image Extraction Pipeline

4.5 Explainability Module

4.5.1 For the 12-Lead Model

Four explainability outputs are generated for the 12-Lead CNN. The first is a Global Grad-CAM Heatmap showing activation across all 12 leads and time steps. The second is the Most Activated Lead, which is the lead with the highest mean Grad-CAM score. The third is a 12-Lead ECG Grid with per-lead Grad-CAM color shading. The fourth is a Combined Saliency Map that averages saliency scores across all leads and shows the result on the lead with the highest mean saliency.

4.5.2 For the Lead II and Image Models

For the Lead II CNN and Image CNN, three outputs are generated: a Grad-CAM overlay using the RdYlBu_r colormap, a Saliency Map using the hot colormap shown as a scatter plot over the signal, and the raw ECG plot with the prediction label and probability annotated on it.

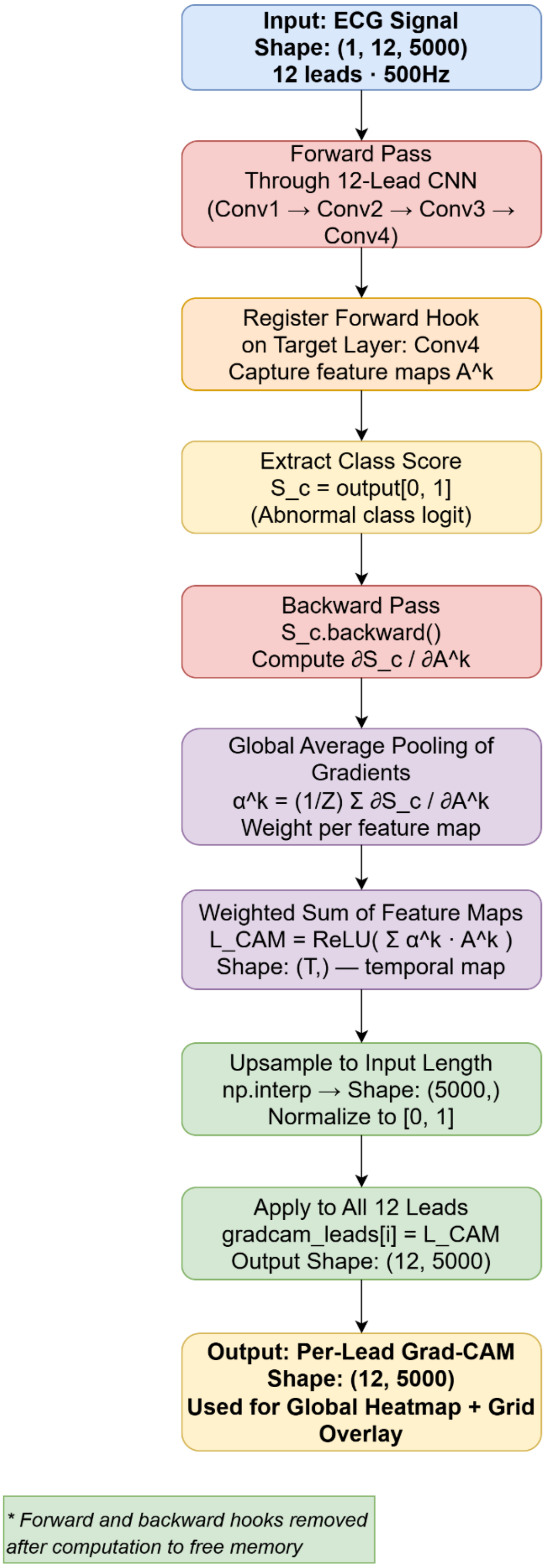


Fig. 4.7 – Grad-Cam Generation for 12-Lead Model.

4.6 UML Diagrams

4.6.1 Use Case Diagram

The use case diagram shows how the user interacts with the system. The user can upload an ECG file, trigger analysis, view the prediction and probability score, view the clinical risk indicator, inspect the explainability visualizations, download the PDF report. All internal steps including routing, preprocessing, inference, and report generation happen automatically inside the application.

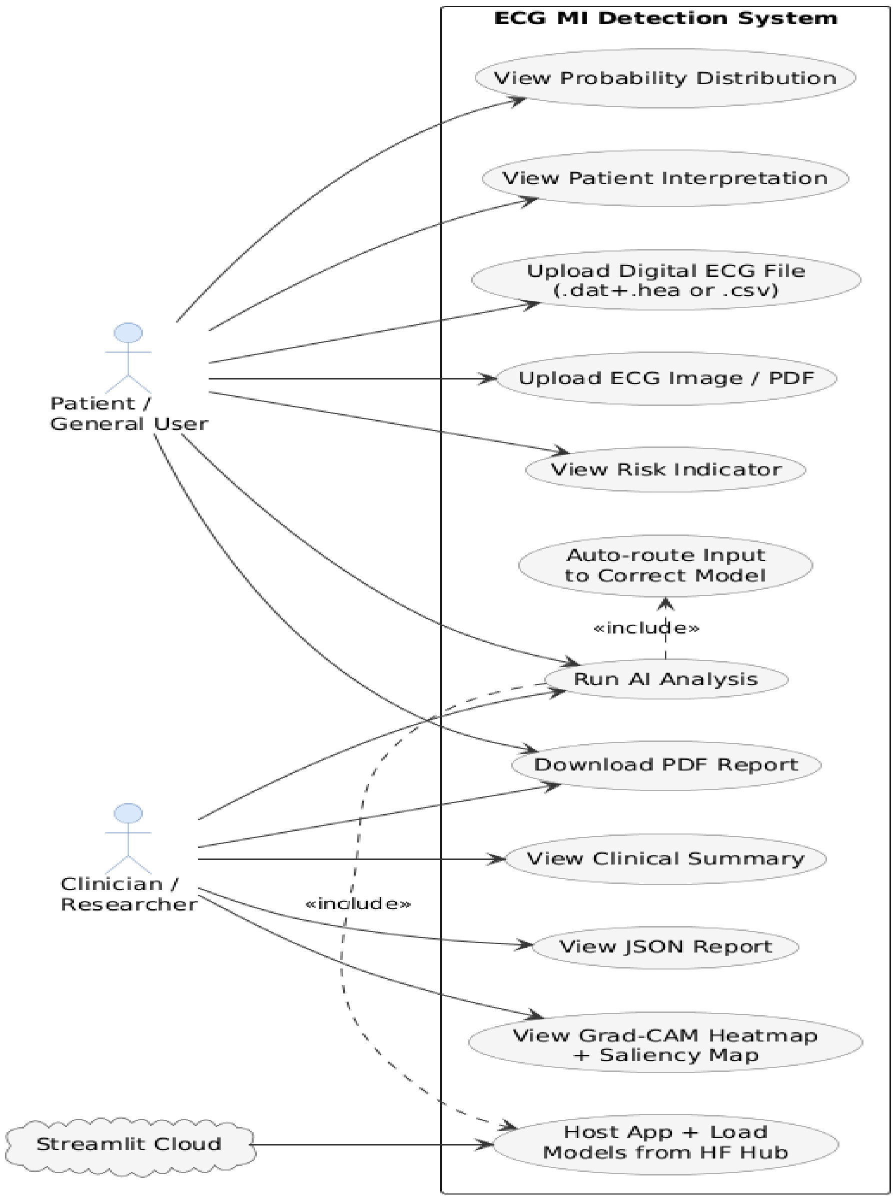


Fig. 4.8 - Use Case Diagram

4.6.2 Sequence Diagram

The sequence diagram shows the order of events when a user uploads a file. The Streamlit app receives the file, the input router checks its type and selects the model, preprocessing runs, CNN inference runs, Grad-CAM and saliency are computed, the PDF report is assembled, and everything is returned to the Streamlit interface for display. In the single-tier architecture there is no HTTP call to a separate backend. All steps happen within the same Python process.

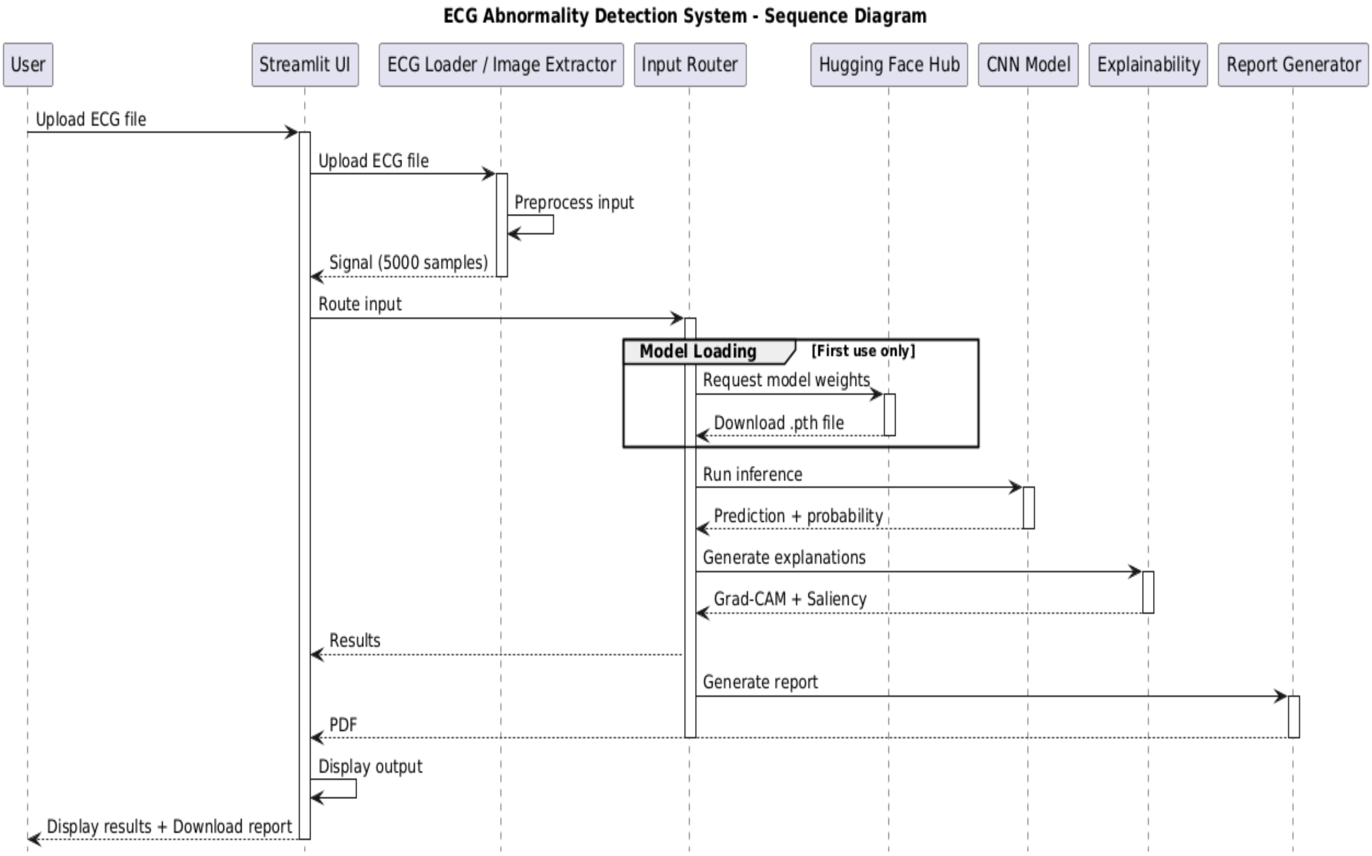


Fig. 4.9 — Sequence Diagram

4.6.3 Activity Diagram

The activity diagram shows the full flow from file upload to report generation, including the decision branches for each input format and the parallel generation of explainability outputs alongside the classification result.

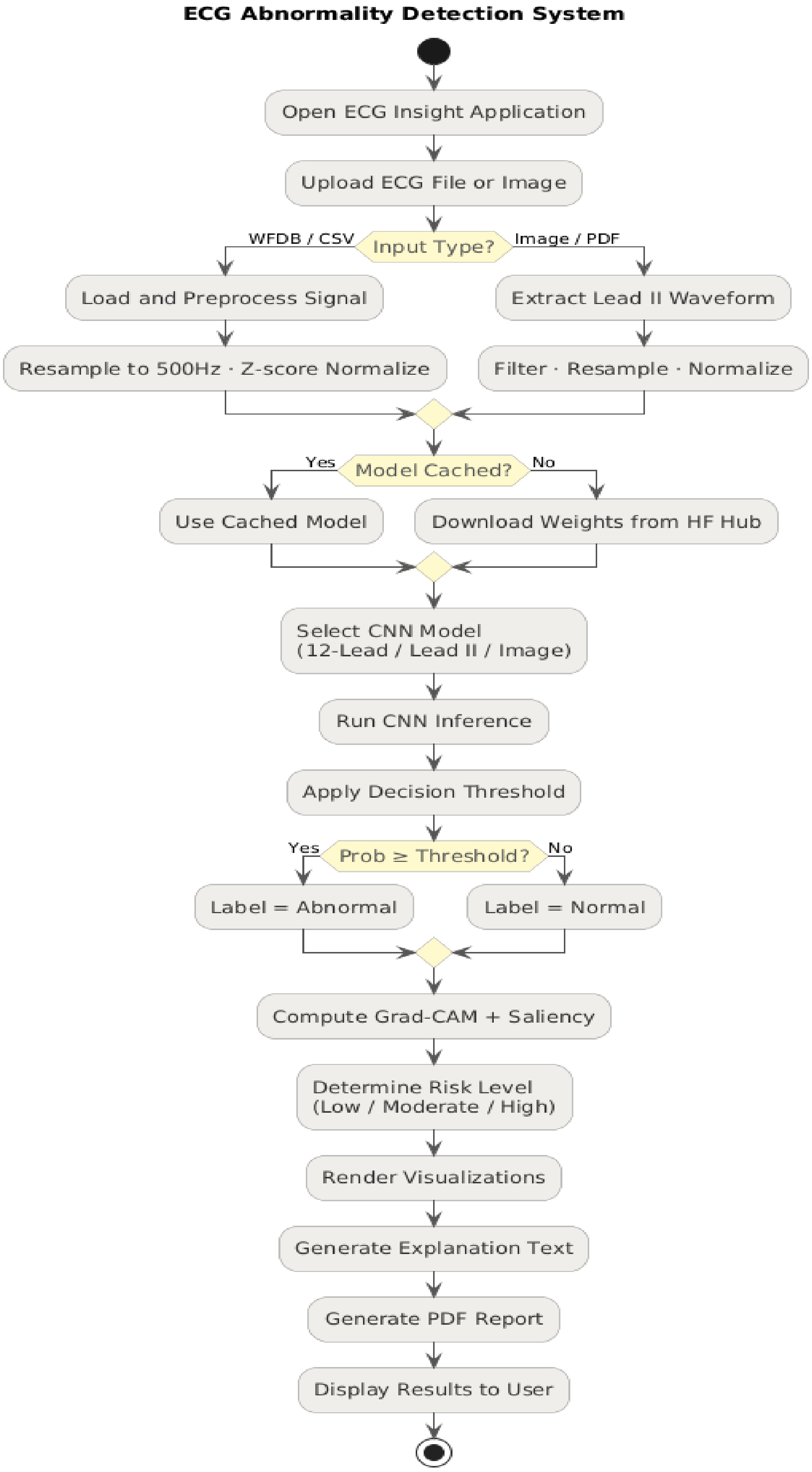


Fig. 4.10 — Activity Diagram

4.6.4 Data Flow Diagram

The data flow diagram shows how data moves through the system at each stage: raw ECG input, normalization, inference, gradient computation, visualization rendering, and final output.

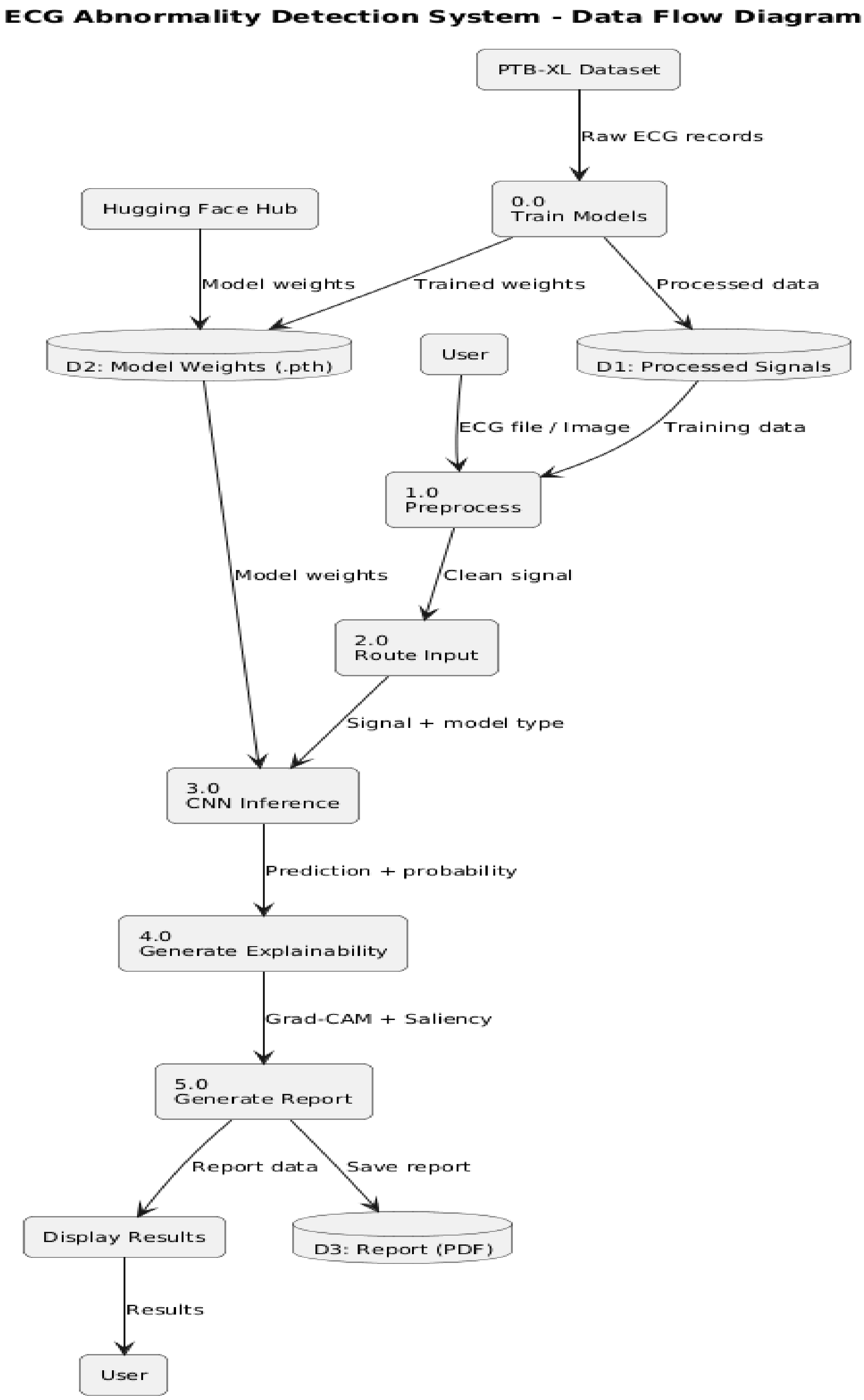


Fig. 4.11 — Data Flow Diagram

5. IMPLEMENTATION

5.1 Dataset

We used the PTB-XL dataset published by Wagner et al. in 2020. Depending on the version downloaded it has 21,799 to 21,837 clinical 12-lead ECG records from 18,869 to 18,885 patients, each 10 seconds long and sampled at 500 Hz giving 5000 samples per lead. The records are labelled with SCP-ECG diagnostic superclasses: normal (NORM), myocardial infarction (MI), ST or T-wave change (STTC), conduction disturbance (CD), and hypertrophy (HYP). We combined MI and STTC into a single positive class representing MI and ischemic abnormalities. All other classes were treated as negative. The data was split 70% training, 15% validation, and 15% test at the patient level. Splitting at the patient level makes sure records from the same patient do not appear in both training and test sets.

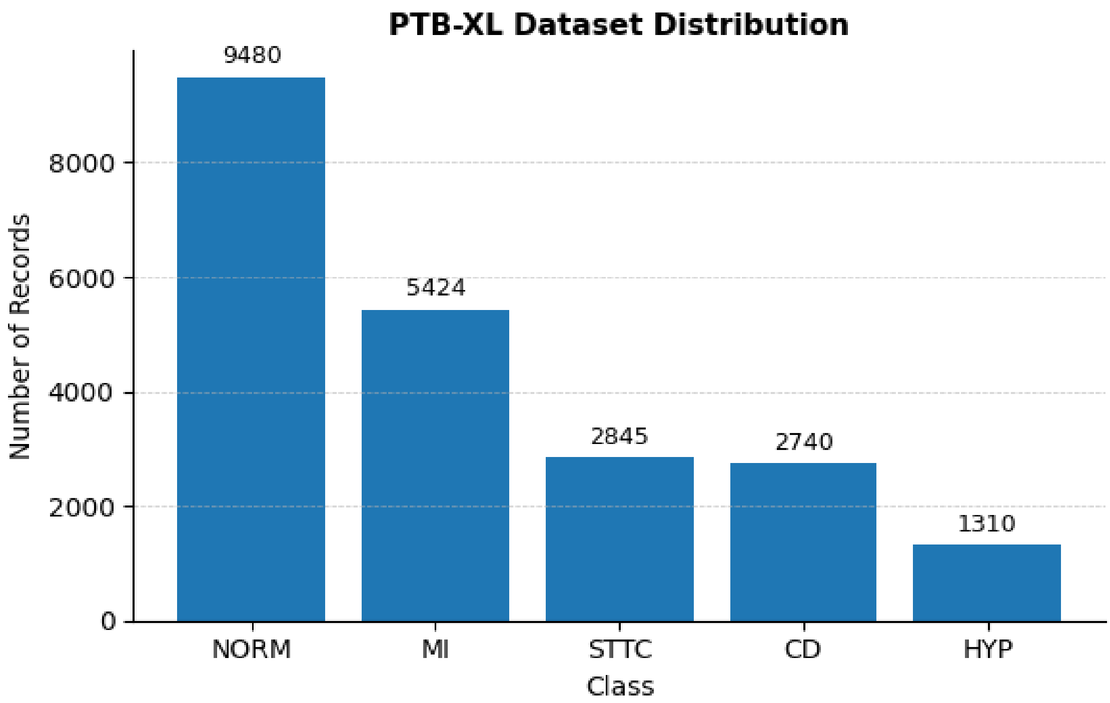


Fig. 5.1 - Dataset Distribution

The PTB-XL dataset we used consists of 21,799 ECG records categorized into five classes. The majority class is normal (NORM), followed by myocardial infarction (MI). This imbalance highlights the need for careful model evaluation.

5.2 Preprocessing

For digital ECG signals, each lead was normalized independently using z-score normalization: (x minus mean) divided by (standard deviation plus 1e-8). No bandpass filter was applied to digital signals since PTB-XL records are already clean clinical recordings. For signals extracted from ECG images, a 4th-order Butterworth bandpass filter from 0.5 to 40 Hz was applied before normalization to remove baseline wander and digitization noise. This filter is only used in the image extraction pipeline and is not part of the main digital ECG preprocessing.

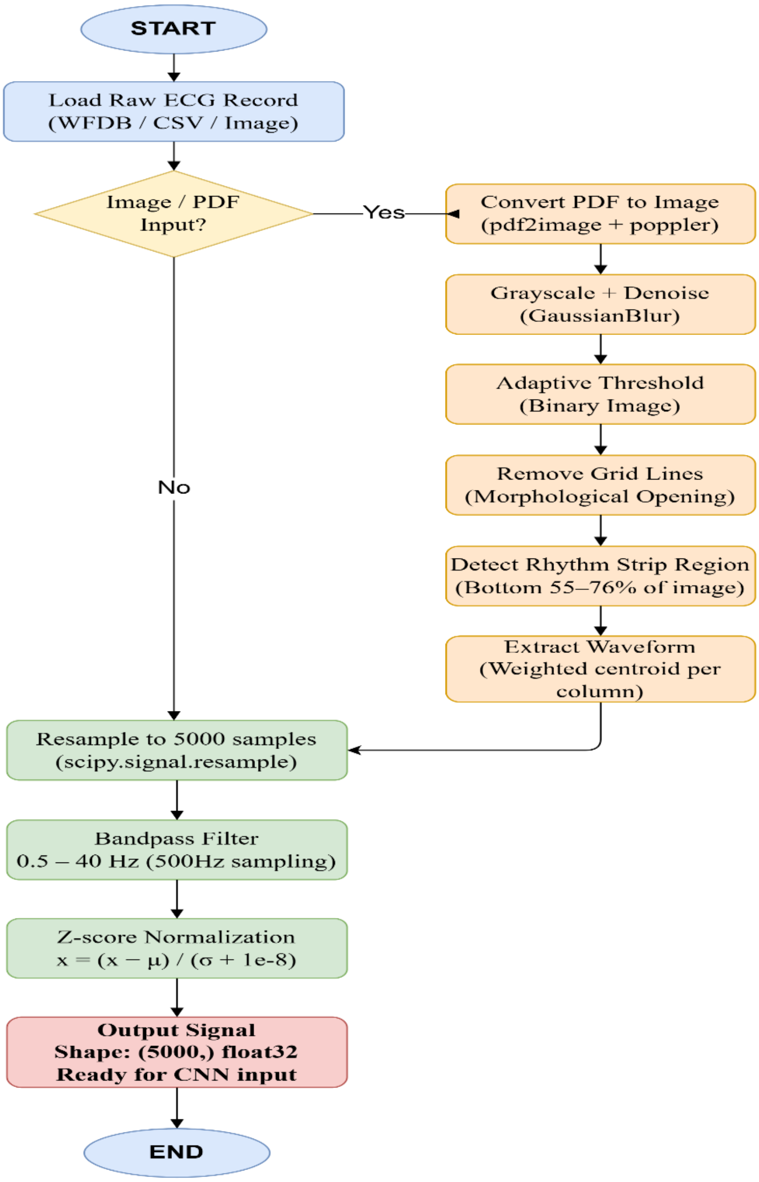


Fig. 5.2 – Signal Preprocessing Pipeline

5.3 Synthetic Image Generation for the Image CNN

No large-scale publicly available ECG image dataset exists with standardized labels matching our schema. The China Physiological Signal Challenge 2018 has incompatible labels. The Ningbo dataset is not publicly available. ECG-Image-Kit by Kshama et al. provides a rendering tool but no labels.

To address this, we generated synthetic ECG images from PTB-XL digital signals using a custom rendering pipeline following the ECG-Image-Kit methodology. Each signal was rendered as a realistic ECG paper image with a pink grid background, Gaussian noise on the signal trace, a small random rotation between minus one and plus one degree to simulate scanning misalignment, and Gaussian blur to simulate compression artifacts. The rhythm strip was then re-extracted from each rendered image using the full image extraction pipeline and used as training input for the Image CNN. This preserves the original PTB-XL labels. The domain gap between these synthetic images and real scanned ECGs from different sources is the main reason the Image CNN performs lower than the digital models.

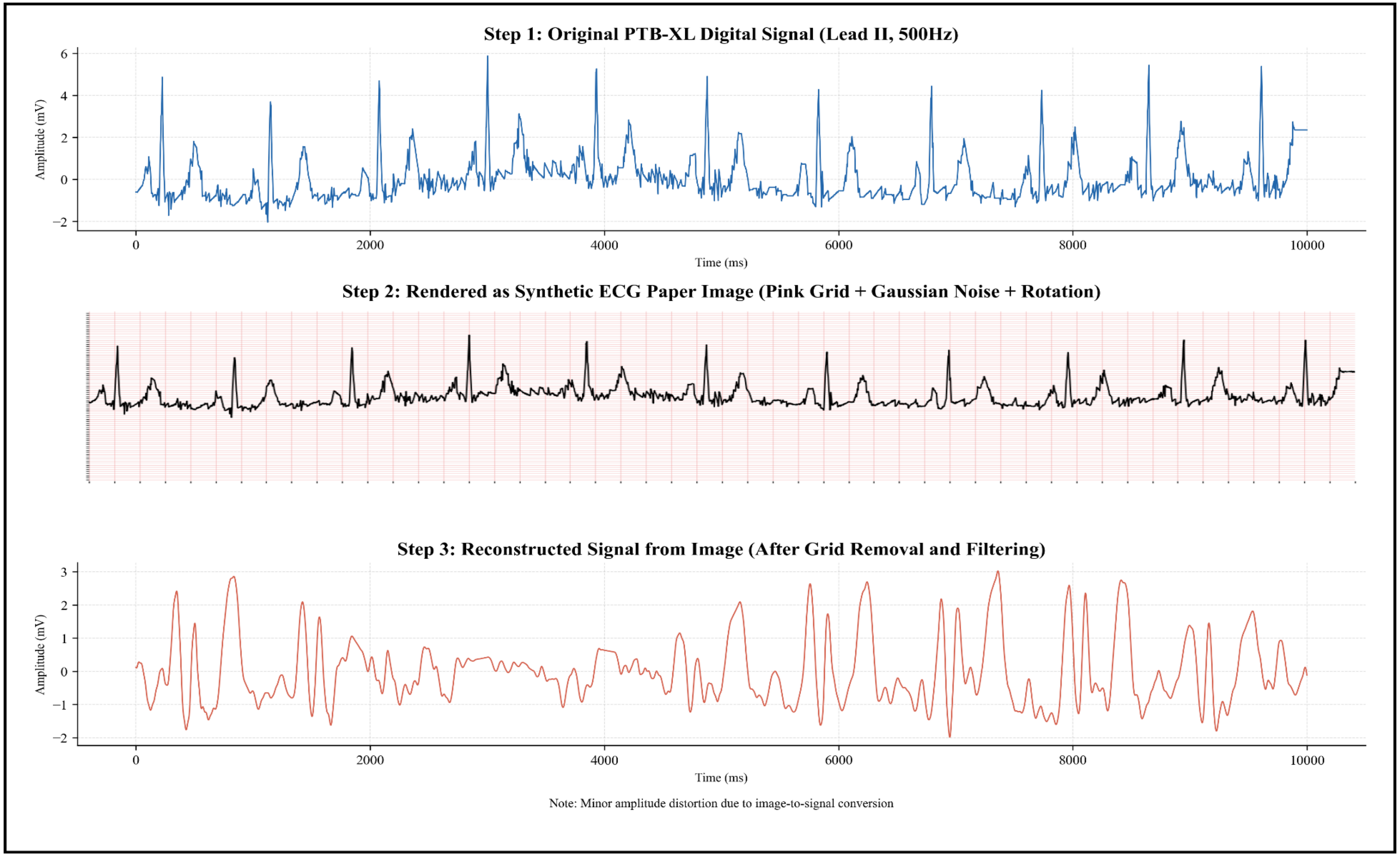


Fig. 5.3 — Synthetic ECG Image Generation Pipeline

5.4 Model Training

All three models were implemented using PyTorch and trained on a CUDA-enabled GPU. The Adam optimizer was employed with a weight decay of 1e-4 and an initial learning rate of 1e-3. A ReduceLROnPlateau scheduler was used to reduce the learning rate by a factor of 0.5 when the validation loss did not improve for five consecutive epochs.

Training was carried out for a maximum of 50 epochs with a batch size of 64, using binary cross-entropy with logits as the loss function. However, early stopping was applied to prevent overfitting, resulting in shorter training durations for all models: the Lead II CNN stopped at epoch 25, the 12-Lead CNN at epoch 38, and the Image CNN at epoch 30.

The classification threshold for each model was tuned on the validation set by selecting the value that gave the highest F1 score. The Lead II CNN threshold was set to 0.48, the 12-Lead CNN to 0.43, and the Image CNN to 0.91. The high threshold for the Image CNN is because the model was overconfident on the training data, which is a common effect when training and test data come from slightly different distributions.

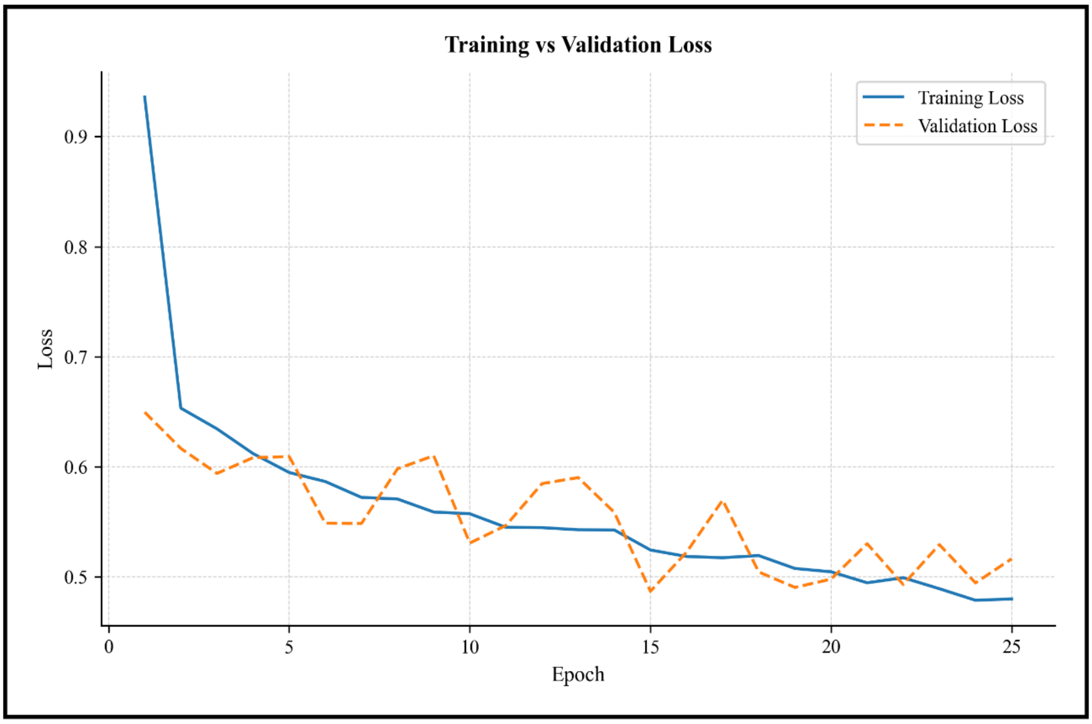


Fig. 5.4 – Lead II CNN: Training and Validation Loss Curves.

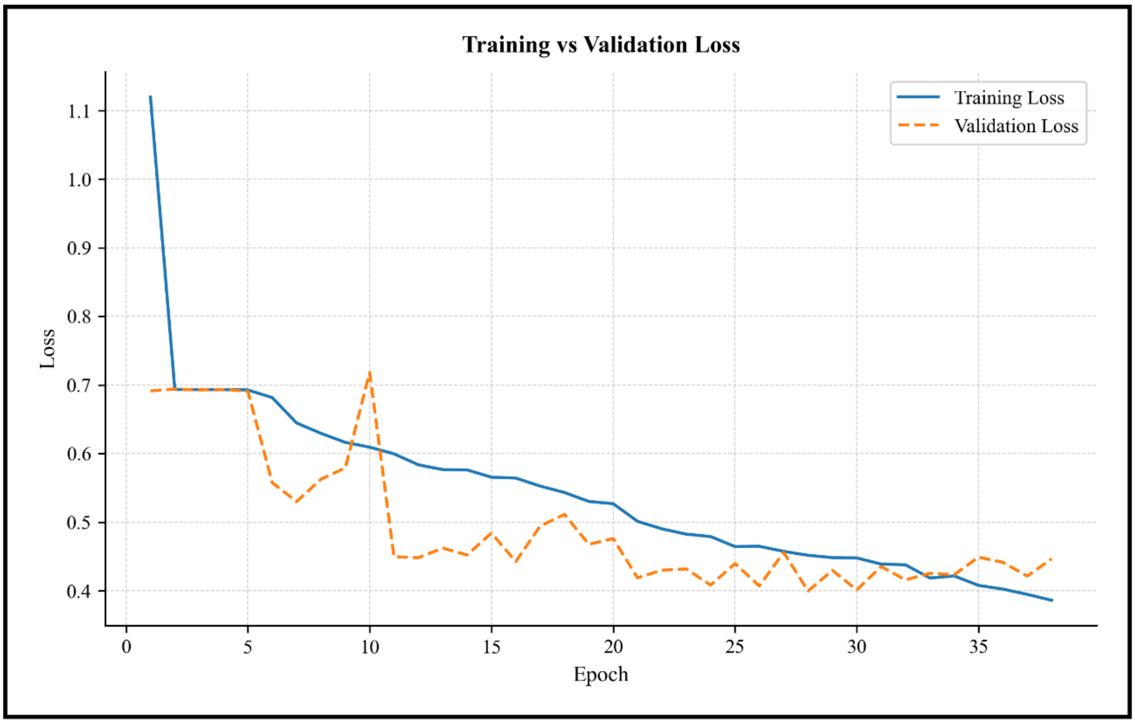


Fig. 5.5 – 12 - Lead CNN: Training and Validation Loss Curves

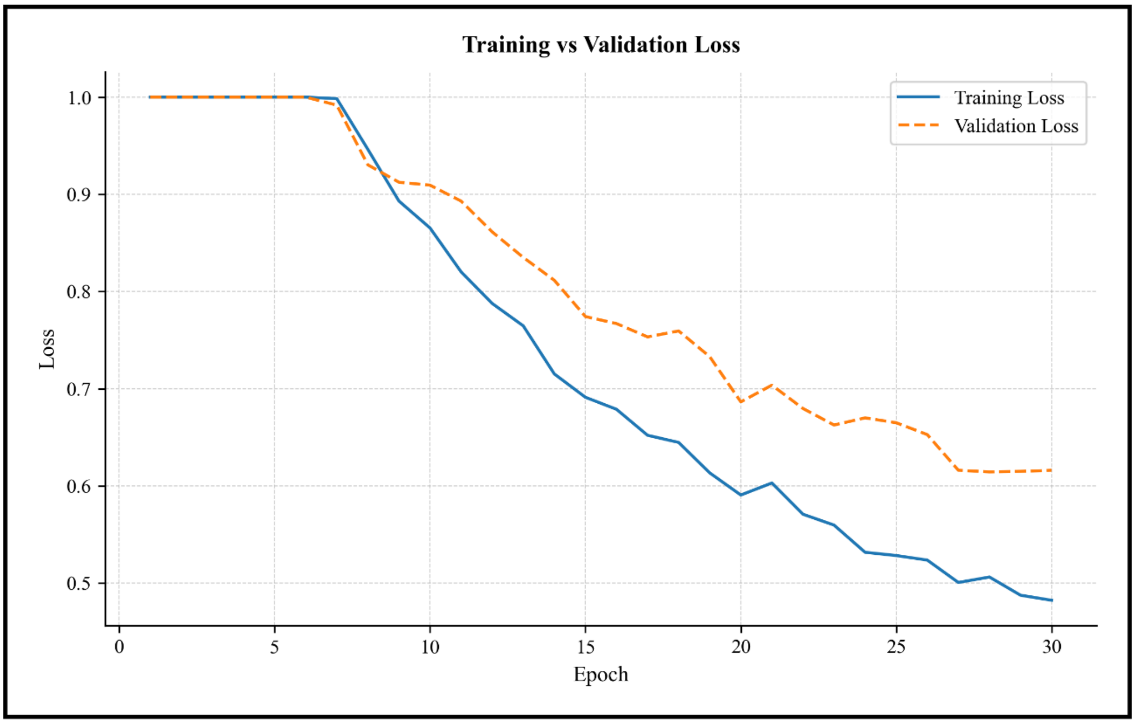


Fig. 5.6 – Image CNN: Training and Validation Loss Curves.

5.5 Deployment

The system is deployed on Streamlit Cloud using a single-tier architecture. All inference functions are called directly from app.py. This is different from a typical two-tier setup where the frontend would send HTTP requests to a separate backend server. That architecture was considered but removed because Streamlit Cloud requires a single-application setup.

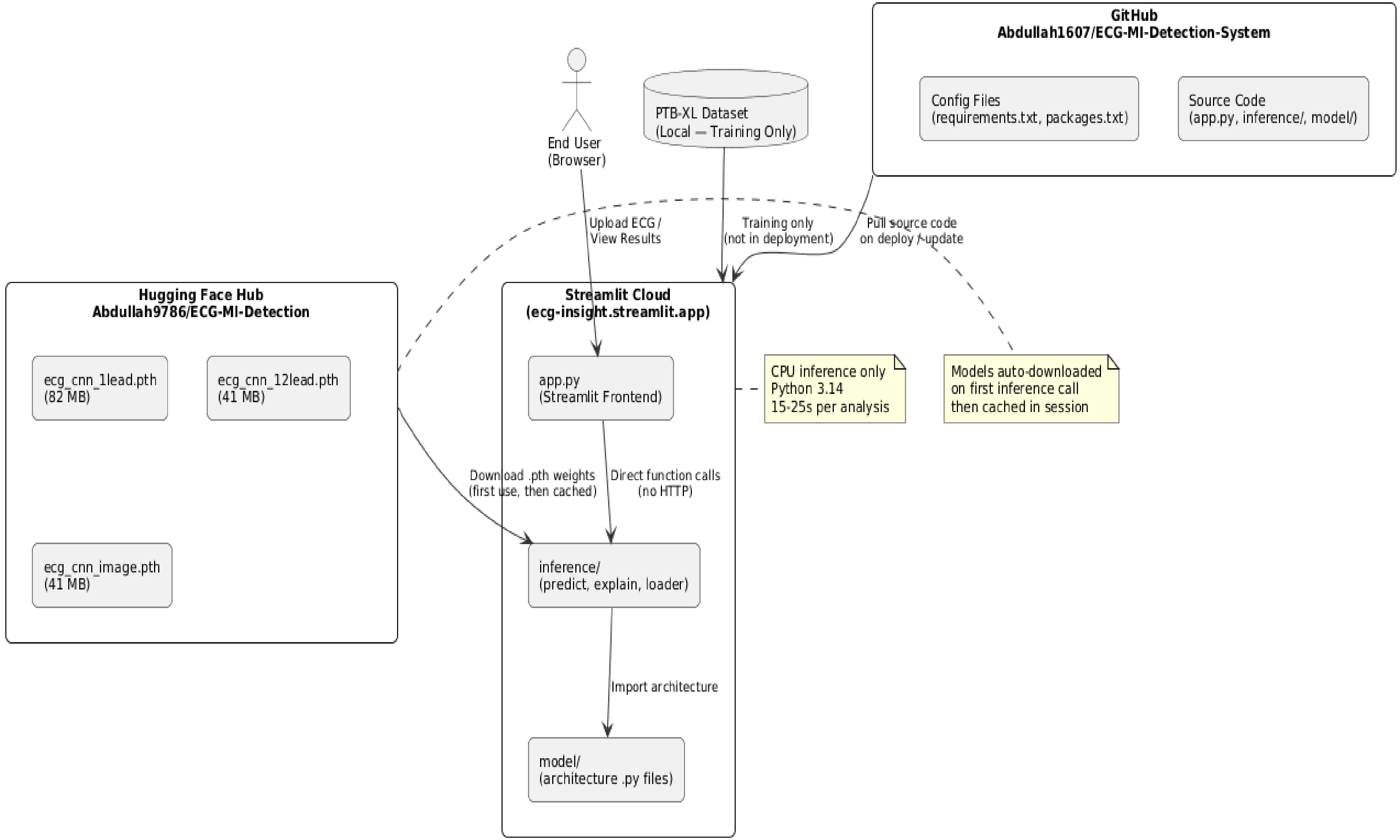
The project folder structure is as follows. app.py is the main Streamlit frontend that also handles all direct inference calls. The inference directory contains all ML inference modules including the ECG loader, preprocessor, image extractor, model runner, Grad-CAM module, saliency module, and report generator. The model directory contains the three trained model weight files: ecg_cnn_leadII.pth, ecg_cnn_12lead.pth, and ecg_cnn_image.pth. The dataset directory contains the ECGDataset class. The training, evaluation, and scripts directories contain training and evaluation code that is not needed for deployment.

The application is deployed from the GitHub repository at
https://github.com/Abdullah1607/ECG-Abnormality-Detection-System.git

The live web application is accessible at
https://ecg-insight.streamlit.app

Streamlit Cloud automatically rebuilds and redeploys the app whenever changes are pushed to the main branch. Inference runs on CPU in the cloud environment and takes approximately 15 to 25 seconds per analysis.

Fig. 5.7 — Streamlit Cloud Deployment Architecture



5.6 Web Application Interface

The Streamlit application has a pink medical aesthetic with a background color of FFF0F3 and a dark red ECG signal color of 8B0000. ECG waveforms are displayed on a simulated ECG paper grid with pink major and minor gridlines. The clinical risk indicator changes color based on the predicted probability: green for low risk below 0.3, orange for moderate risk between 0.3 and 0.6, and red for high risk above 0.6. Probability bars show the exact percentage for each class.

There are two interpretation tabs for every prediction. The first tab is written in plain language for the patient and explains the result without technical terms. The second tab uses clinical terminology and is intended for the doctor. For image-based inputs, a research note is shown that explains the synthetic training approach and marks the result as experimental.

Users can download a PDF report with all visualizations embedded. The PDF includes the ECG plot, classification result, probability scores, Grad-CAM heatmaps, and saliency maps.

5.7 Ablation Study

We ran an ablation study to check that our main design decisions were correct. We looked at three factors. For sampling frequency, models trained on 500 Hz signals outperformed 100 Hz models by about 0.038 AUC, confirming that higher frequency captures clinically important signal details. For dataset size, models trained on only 10% of the data showed AUC reductions of about 0.078, showing that the full dataset size is necessary. For compute, GPU and CPU training gave the same final results but GPU training cut per-epoch time from about 20 minutes to under 2 minutes.

6. TESTING

6.1 System Testing Overview

We tested the system at three levels: unit testing for individual functions, integration testing to check that all modules work together, and system testing to verify the full deployed application. All test cases were run manually on the locally hosted application using ECG files from the PTB-XL test set and real scanned ECG images.

6.2 Unit Testing

Table 6.1: Unit Test Cases

TC No	Test Case	Input	Expected Output	Result
UT-01	Z-score normalization	Lead array with known mean and std	Normalized array with mean near 0 and std near 1	PASS
UT-02	WFDB file loading	Valid .dat and .hea file pair	12 leads each with 5000 samples	PASS
UT-03	CSV single-column read	1-column CSV file	Lead II tensor of shape (1, 5000)	PASS
UT-04	CSV 12-column read	12-column CSV file	12-lead tensor of shape (12, 5000)	PASS
UT-05	Input router for WFDB	WFDB file	Routes to 12-Lead CNN	PASS
UT-06	Input router for 1-column CSV	Single-column CSV	Routes to Lead II CNN	PASS
UT-07	Input router for image	PNG file	Routes to Image CNN pipeline	PASS
UT-08	Lead II CNN inference	Valid Lead II tensor	Float probability between 0 and 1	PASS
UT-09	12-Lead CNN inference	Valid 12-lead tensor	Float probability between 0 and 1	PASS
UT-10	Image extraction pipeline	ECG paper image	Extracted waveform of 5000 samples	PASS
UT-11	Image CNN inference	Extracted waveform tensor	Float probability between 0 and 1	PASS
UT-12	Grad-CAM for 12-Lead	12-lead tensor and model weights	Heatmap of shape (12, 5000)	PASS
UT-13	Saliency map generation	Any valid tensor	Gradient array same shape as input	PASS
UT-14	PDF report generation	Prediction results and plots	Valid PDF file with embedded figures	PASS

6.3 Integration Testing

Table 6.2: Integration Test Cases

TC No	Test Case	Input	Expected Output	Result
IT-01	End to end WFDB upload	WFDB file via Streamlit UI	Classification, Grad-CAM, and PDF rendered	PASS
IT-02	End to end 12-column CSV	12-column CSV via UI	Classification and all visualizations rendered	PASS
IT-03	End to end 1-column CSV	Single-column CSV via UI	Lead II classification result rendered	PASS
IT-04	Image upload and Image CNN	PNG ECG image via UI	Image extraction runs, Image CNN result shown with research note	PASS
IT-05	PDF report download	Any valid ECG classification	PDF with embedded visualizations downloads correctly	PASS
IT-06	Risk indicator display	MI positive ECG	Red risk indicator shown	PASS
IT-07	Patient and clinician tabs	Any prediction result	Both interpretation tabs render correctly	PASS
IT-08	Streamlit Cloud deployment	GitHub push to main branch	App rebuilds and runs on Streamlit Cloud	PASS

6.4 System Testing

Table 6.3: System Test Cases

TC No	Test Case	Input	Expected Output	Result
ST-01	Unsupported file type	Text file upload	Error message displayed with no crash	PASS
ST-02	Corrupted file handling	Malformed CSV	Graceful error message returned	PASS
ST-03	Poor quality ECG image	Low resolution scan	Image extraction runs, result shown as experimental	PASS
ST-04	Threshold application	MI ECG near decision boundary	Correct label given threshold 0.48 or 0.43 or 0.91	PASS
ST-05	CPU inference timing on cloud	WFDB upload on Streamlit Cloud	Result returned within 15 to 25 seconds	PASS
ST-06	Acceptance testing	Full workflow by a non-technical user	All features work correctly from upload to report download	PASS

7. RESULTS

7.1 Model Performance

All three models were evaluated on the held-out test set of approximately 3,275 patient records. Table 7.1 shows the complete results. The thresholds were chosen by maximizing F1 score on the validation set.

Table 7.1: Model Performance Comparison

Metric	Lead II CNN	12-Lead CNN	Image CNN
ROC-AUC	0.8528	0.8991	0.8083
Accuracy	0.7682	0.8099	~0.75
Sensitivity	0.8500	0.8674	0.8260
Specificity	0.6634	0.7580	0.6180
F1 Score	0.8045	0.8436	0.7800
Threshold	0.48	0.43	0.91

The 12-Lead CNN gives the best results across all metrics. It outperforms the Lead II CNN by 0.047 AUC and 0.095 specificity. Higher specificity means fewer false positives, which is important in a clinical setting because unnecessary interventions carry their own risks. Both digital models were tuned to prioritize sensitivity since missing a real MI case is more dangerous than a false alarm.

The Image CNN got an AUC of 0.808 and sensitivity of 0.826. These numbers are lower than the digital models, which is expected since the model was trained on synthetic images. The threshold of 0.91 is much higher than the other models because the Image CNN was overconfident on the training data. This is a common effect when the training and test distributions differ slightly, which is the case here because the test images are also synthetic but from a different rendering pass. The image pathway is clearly marked as experimental in the application.

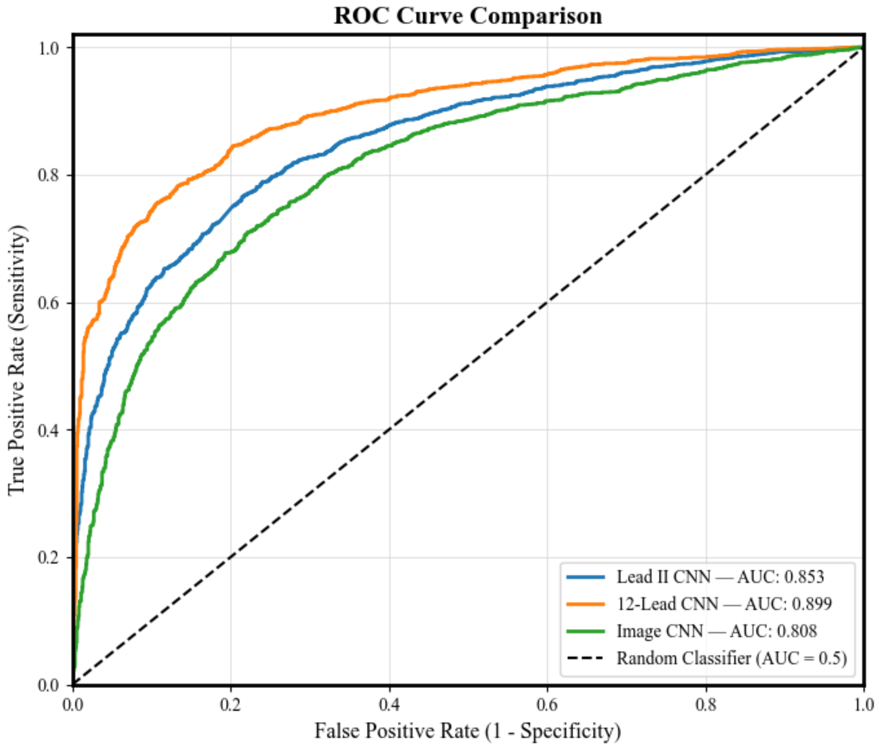


Fig. 7.1 – ROC Curve Comparison Across All Three Models.

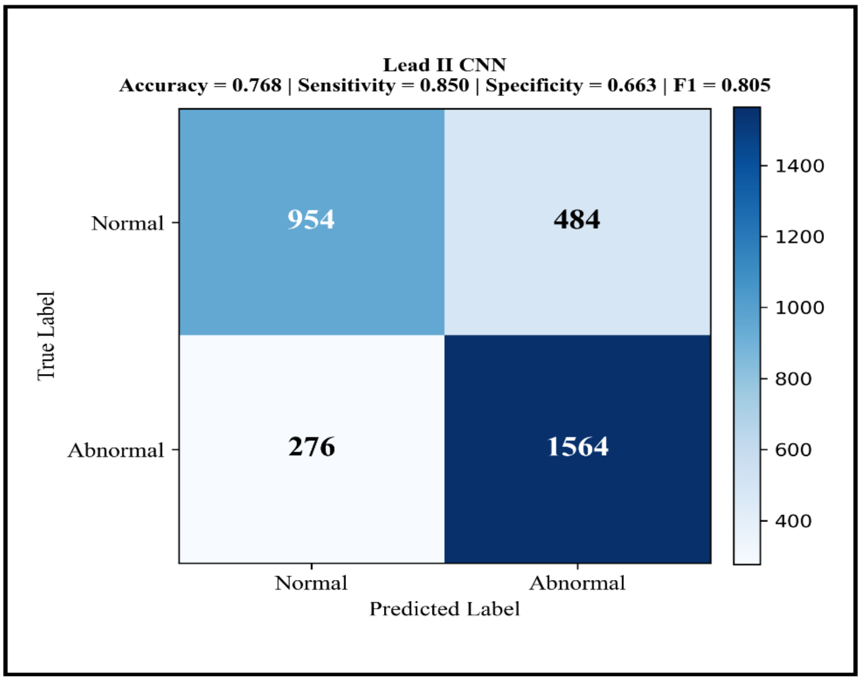


Fig. 7.2 – Confusion Matrix: Lead II CNN

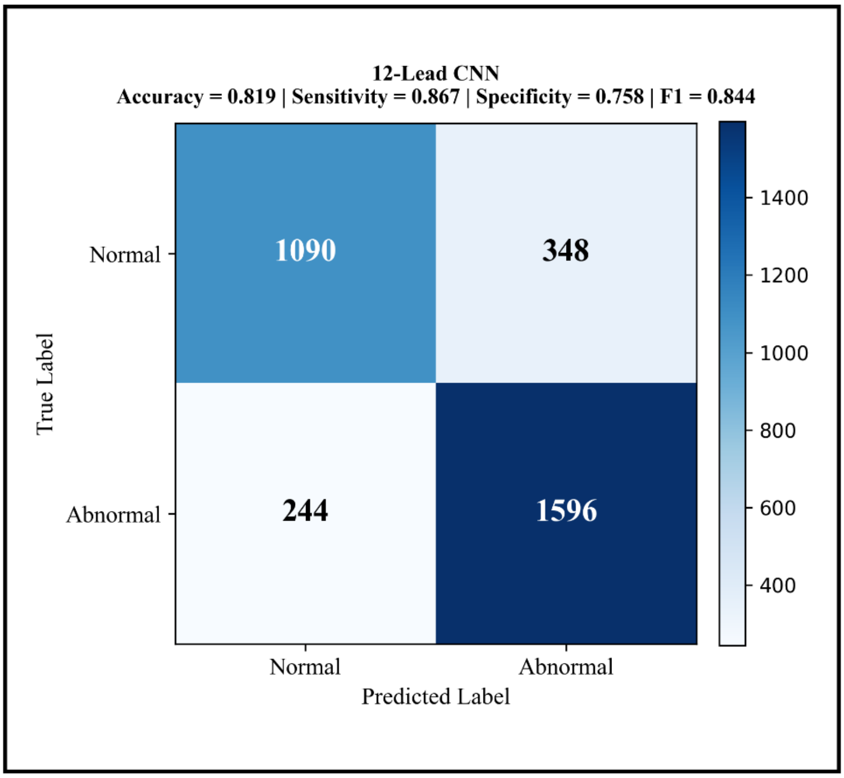


Fig. 7.3 – Confusion Matrix:12-Lead CNN

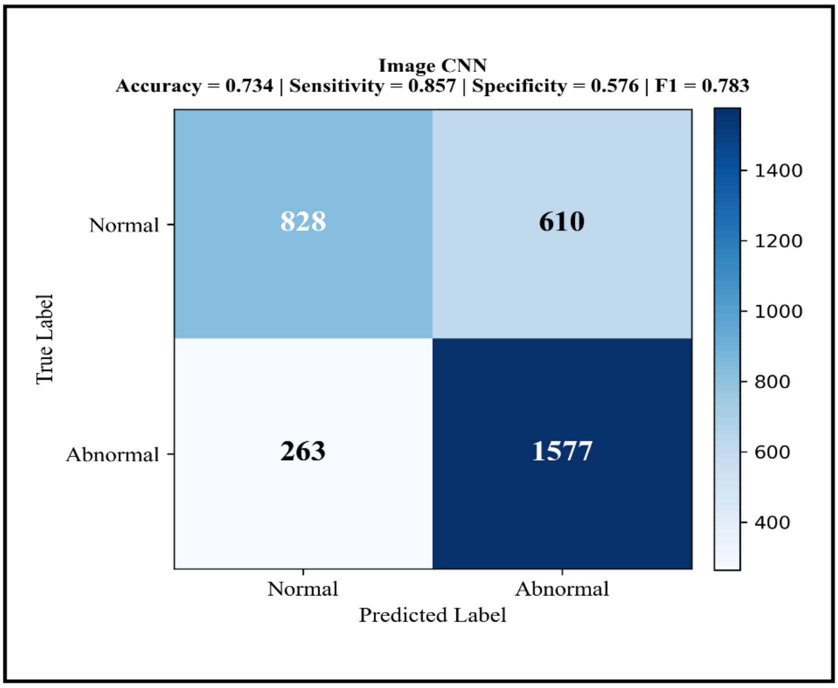


Fig. 7.4 – Confusion Matrix: Image CNN

7.2 Explainability Outputs

The Grad-CAM heatmaps for the 12-Lead CNN consistently activated over the ST-segment and T-wave regions in leads II, V4, V5, and V6 for positive MI cases. These are the exact leads and waveform regions that cardiologists look at first when checking for ischemia. For inferior MI cases the highest activation was in leads II, III, and aVF, which matches the established clinical pattern for that MI subtype. This alignment with known clinical markers gives us reasonable confidence that the model learned the right features rather than dataset artifacts.

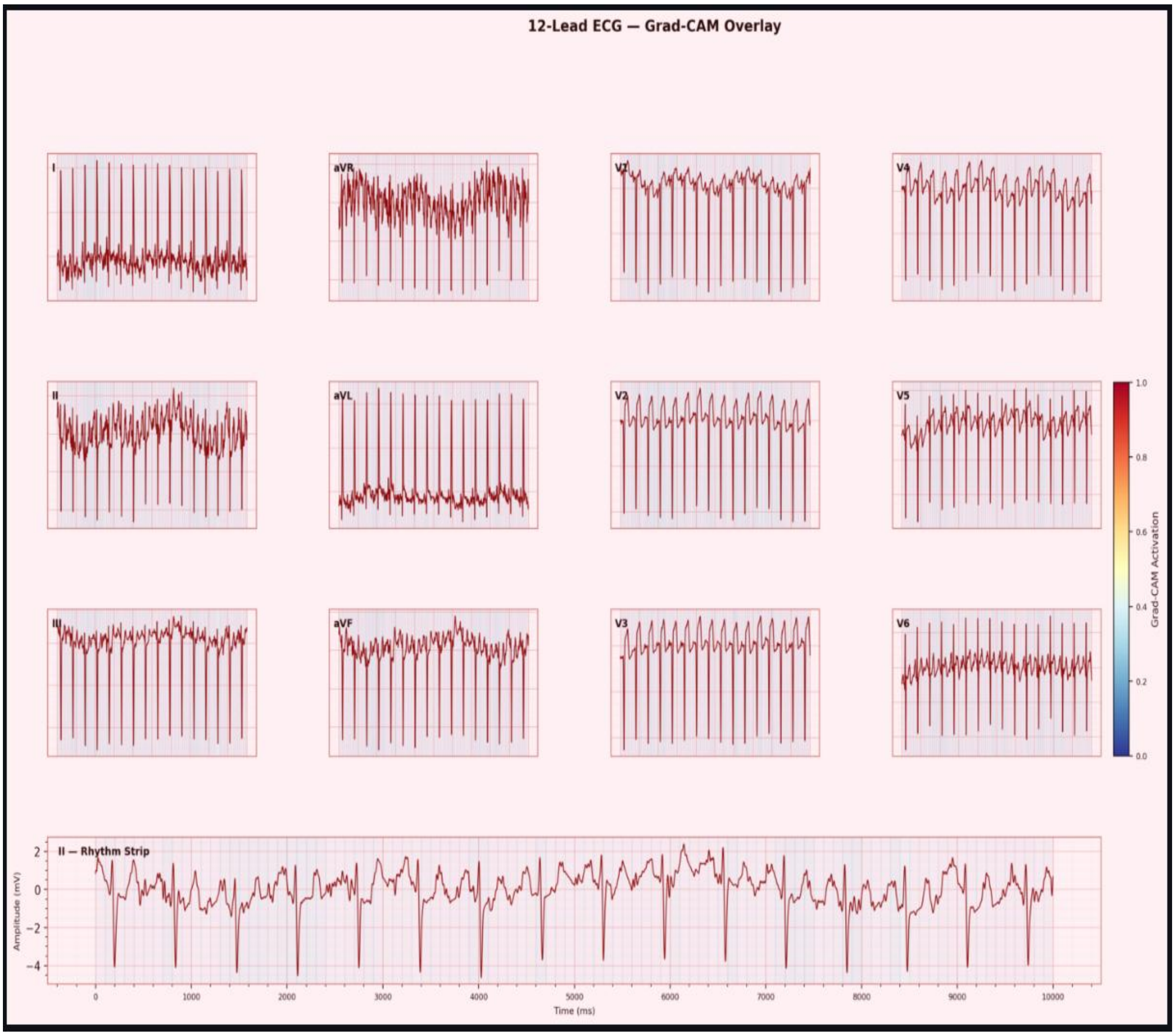


Fig. 7.5 - Grad-CAM on 12-Lead Grid

The saliency maps showed high activation around the QRS complex onset and the ST-segment transition region. This is consistent with where the most diagnostically important information is in an ECG signal.

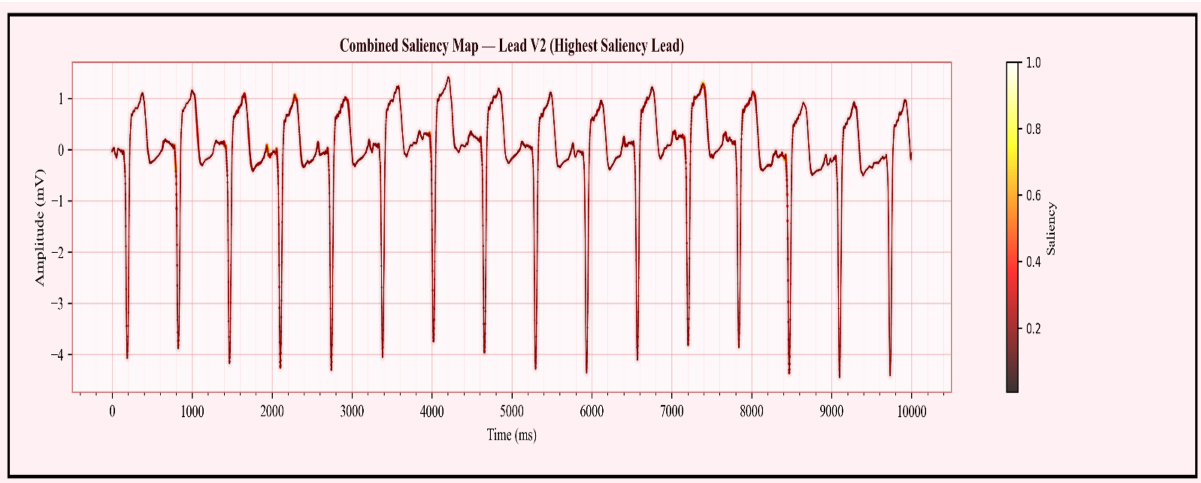


Fig. 7.6 – Saliency Map on Lead II CNN

7.3 Web Application

The deployed Streamlit Cloud application worked correctly for all tested input formats. The inference time on Streamlit Cloud CPU was approximately 15 to 25 seconds per analysis including Grad-CAM computation and PDF generation. This is acceptable for a screening tool where the result does not need to appear in real time. All UI features including the clinical risk indicator, probability bars, interpretation tabs, PDF download.

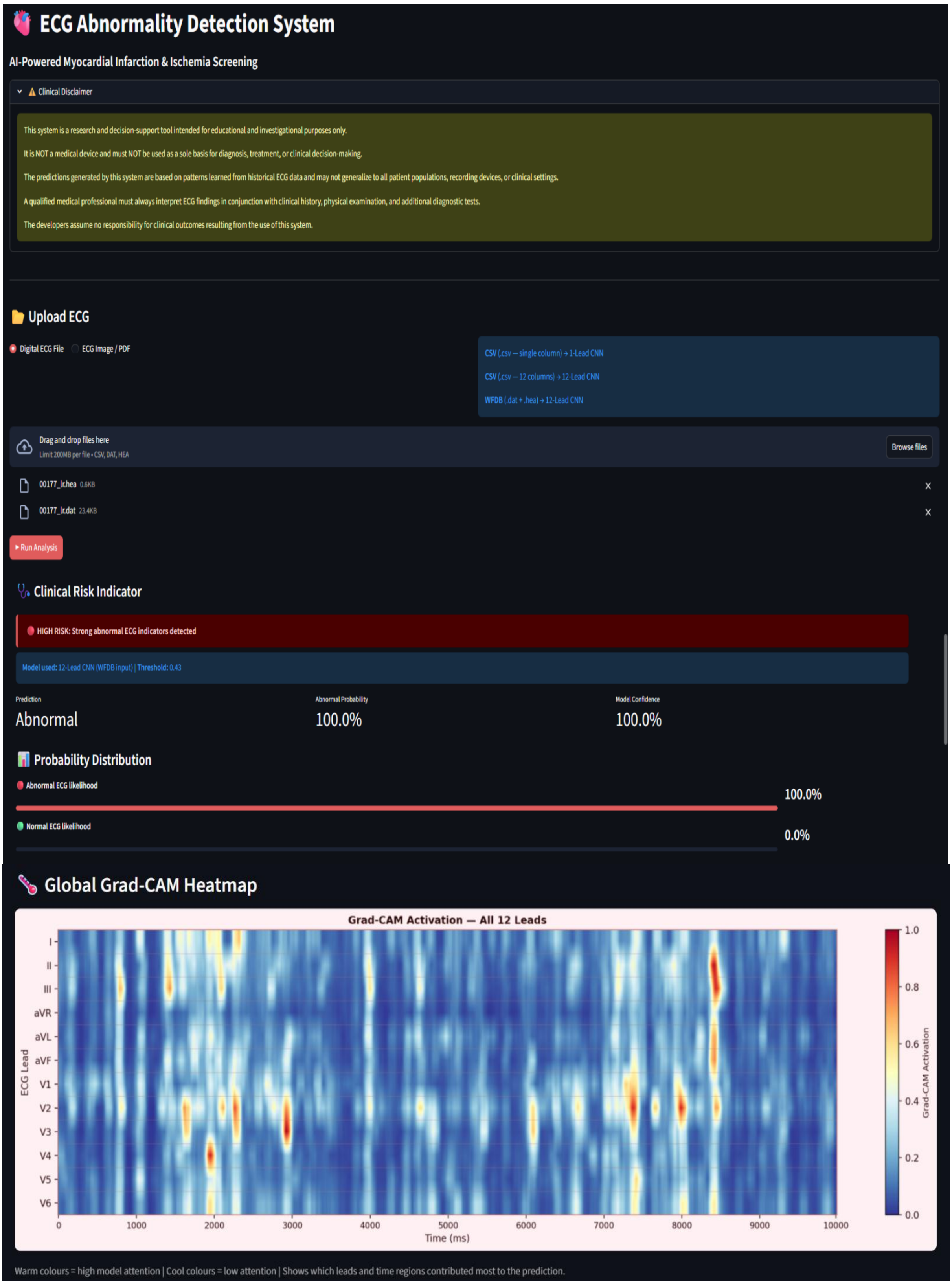


Fig. 7.7 - Application Interface

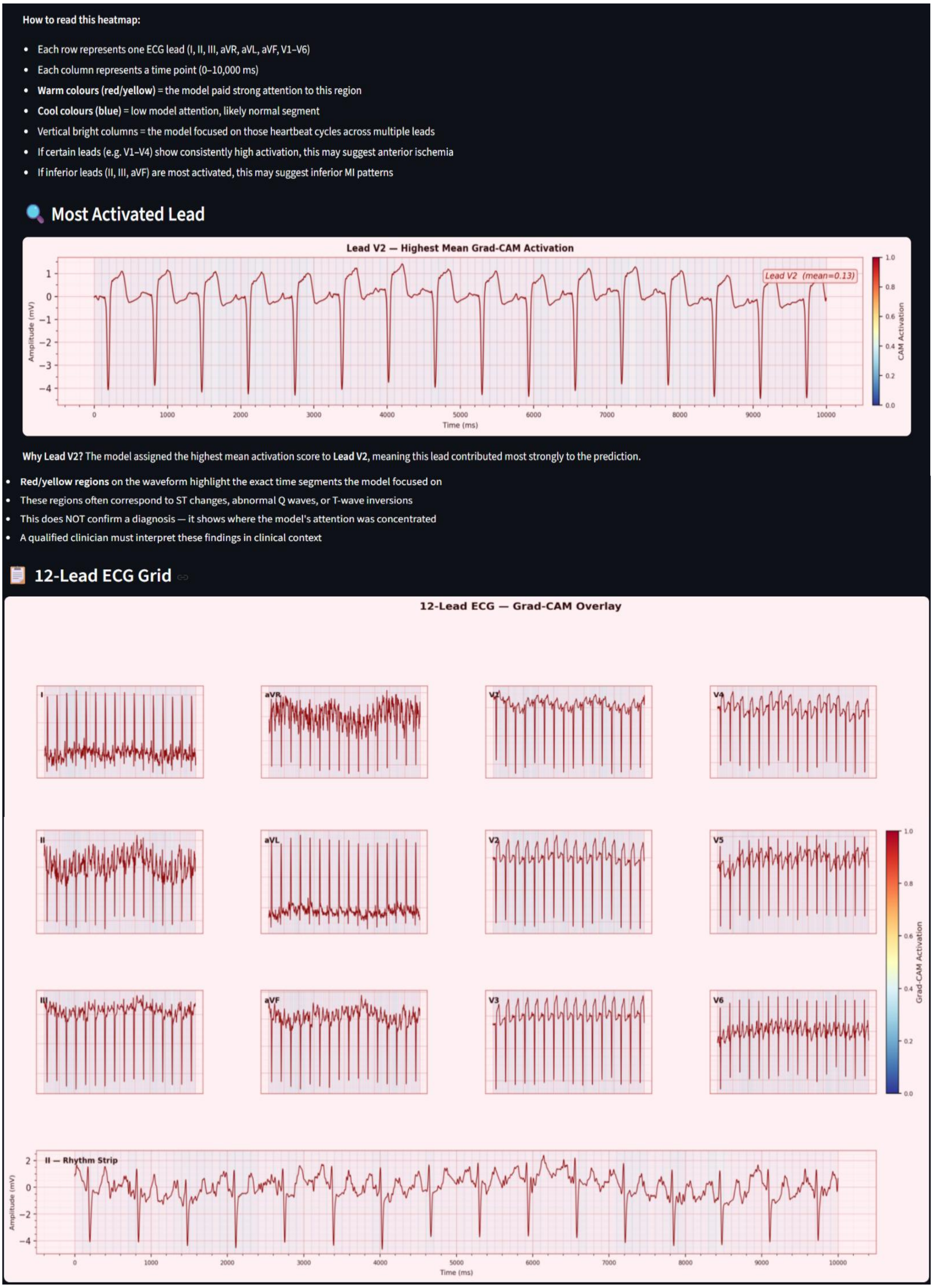


Fig. 7.7 - Application Interface

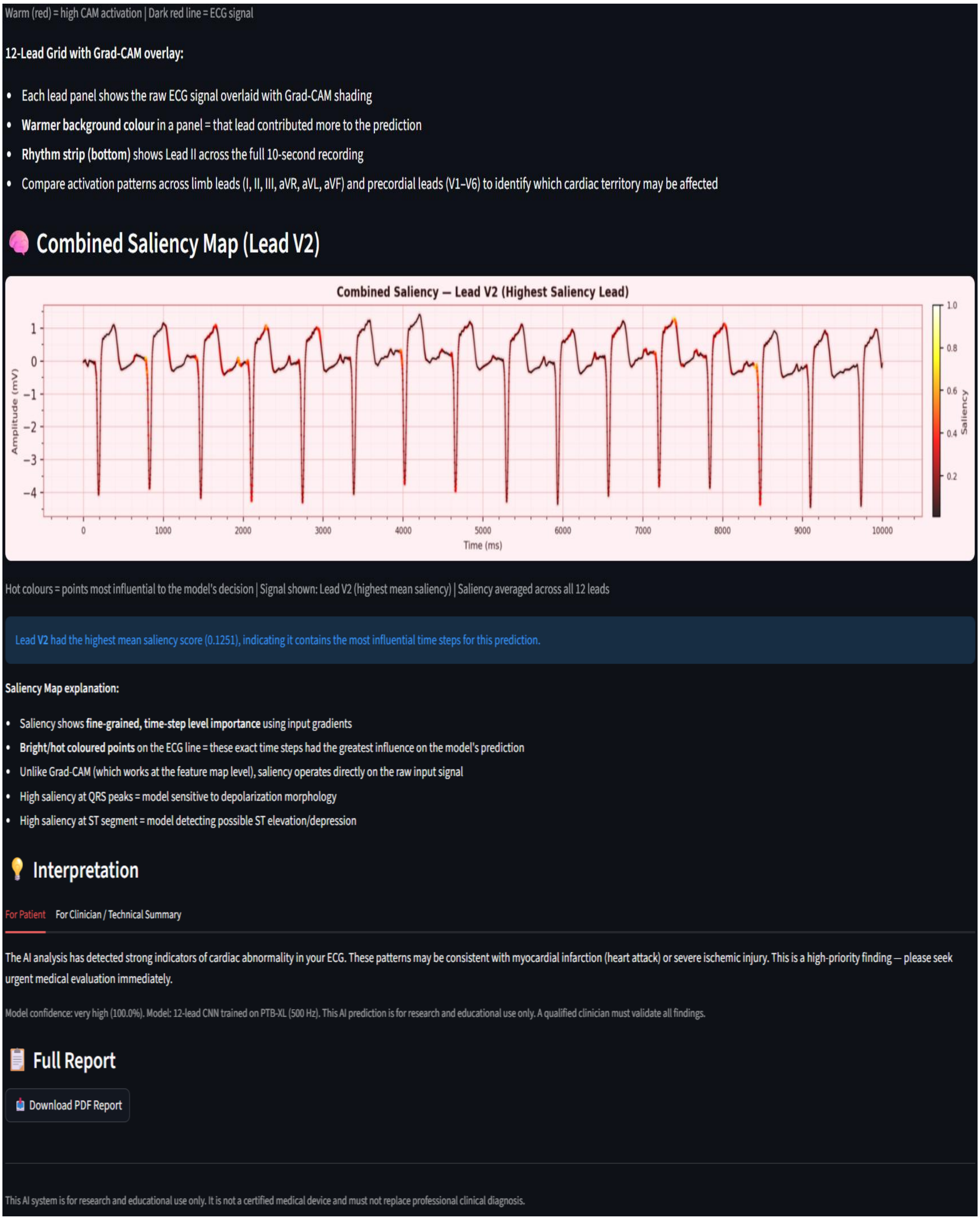


Fig. 7.7 - Application Interface

7.4 Ablation Study Results

Table 7.2: Ablation Study Results (12-Lead CNN)

Configuration	ROC-AUC	F1 Score	Observation
500 Hz, full dataset, GPU	0.8991	0.8436	Final deployed configuration
100 Hz, full dataset, GPU	0.8612	0.8101	AUC dropped by 0.038
500 Hz, 10% dataset, GPU	0.8213	0.7734	AUC dropped by 0.078
500 Hz, full dataset, CPU	0.8991	0.8436	Same result but training took about 10 times longer

8. CONCLUSION AND FUTURE WORKS

8.1 Conclusion

This project built a complete system for detecting myocardial infarction and ischemic abnormalities from ECG signals. We trained three models, added Grad-CAM and saliency map explainability, and deployed everything as a working Streamlit Cloud application that anyone can access. The goal was to make something that is not just accurate but also usable and understandable.

The 12-Lead CNN got a ROC-AUC of 0.8991 and F1 of 0.8436 on the PTB-XL test set. This is competitive with published systems that only do detection without any explainability or deployment. The Lead II CNN with AUC 0.8528 is a lighter option when 12-lead data is not available. The Image CNN with AUC 0.8083 is a first step toward supporting scanned ECG inputs in the same pipeline.

The Grad-CAM outputs consistently activated over ST-segment and T-wave regions in the right leads for each MI subtype. This shows the model is attending to clinically relevant features and not just memorizing patterns in the data. The Streamlit Cloud deployment with single-tier architecture, automatic input routing, PDF report, and patient and clinician interpretation tabs makes the system practically accessible without needing any special software.

There are some honest limitations to acknowledge. The Image CNN was trained on synthetic data and may not perform as well on real scanned ECGs from diverse sources. The Lead II CNN has a specificity of only 0.66 which means it produces more false positives than the 12-lead model. Borderline records in PTB-XL labelled at low confidence are difficult for any model to get right. Inference on Streamlit Cloud CPU takes 15 to 25 seconds, which is fine for screening but not for real-time use.

8.2 Future Works

The most important next step for the image pathway is collecting or building a real-world labelled ECG image dataset with standardized labels. This would close the domain gap and make the Image CNN more reliable on actual clinical scans.

Extending the system to detect more cardiac conditions beyond MI and ischemia, such as arrhythmias and conduction disorders, would make it more useful in a general clinical setting.

Using a transformer-based architecture with attention mechanisms would improve both performance and explainability. Attention weights provide a more natural and built-in form of explanation compared to post-hoc Grad-CAM.

Testing the system on ECG data from Indian hospitals would tell us how well it generalizes to local patient populations and recording equipment, which may differ from the European PTB-XL cohort.

Finally, adding real-time ECG streaming support would make the system suitable for continuous monitoring applications, though this would require significant optimization of the inference pipeline.

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10. APPENDIX

A. Sample Code

A.1 Z-Score Normalization

```
def preprocess_signal(signal):
    mean = np.mean(signal, axis=-1, keepdims=True)
    std  = np.std(signal, axis=-1, keepdims=True) + 1e-8
    return (signal - mean) / std
```

A.2 Input Routing Logic

```
def route_input(file):
    ext = file.name.split('.')[-1].lower()
    if ext in ['dat','hea']:
        return load_wfdb(file), '12lead'
    if ext == 'csv':
        df = pd.read_csv(file)
        if df.shape[1] == 12: return tensor(df.values.T), '12lead'
        if df.shape[1] == 1:  return tensor(df.values.T), 'lead2'
    if ext in ['png','jpg','jpeg','pdf']:
        return extract_from_image(file), 'image'
    raise ValueError('Unsupported file type')
```

A.3 12-Lead CNN Architecture (PyTorch)

```
class ECGNet12Lead(nn.Module):
    def __init__(self):
        super().__init__()
        self.conv1 = nn.Sequential(nn.Conv1d(12,32,7,padding=3),
            nn.BatchNorm1d(32), nn.ReLU(), nn.MaxPool1d(2),
nn.Dropout(0.3))
        self.conv2 = nn.Sequential(nn.Conv1d(32,64,5,padding=2),
            nn.BatchNorm1d(64), nn.ReLU(), nn.MaxPool1d(2),
nn.Dropout(0.3))
        self.conv3 = nn.Sequential(nn.Conv1d(64,128,5,padding=2),
            nn.BatchNorm1d(128), nn.ReLU(), nn.MaxPool1d(2))
        self.conv4 = nn.Sequential(nn.Conv1d(128,128,3,padding=1),
            nn.BatchNorm1d(128), nn.ReLU(), nn.MaxPool1d(2))
        self.fc1 = nn.Linear(128*312, 256)
        self.dropout = nn.Dropout(0.3)
        self.fc2 = nn.Linear(256, 1)
```

B. Hardware and Software Requirements

B.1 Hardware Requirements

The system was developed and tested on standard computing hardware. A minimum configuration is required for basic functionality, while higher specifications improve training performance.

- **Processor:** Intel Core i5 (8th Generation) or AMD Ryzen 5 or higher
- **RAM:** Minimum 8 GB (16 GB recommended for model training)
- **GPU:** NVIDIA CUDA-enabled GPU recommended for training; CPU is sufficient for inference and Streamlit deployment
- **Storage:** At least 20 GB of free space for dataset storage and model weight files
- **Operating System:** Windows 10/11, Ubuntu 20.04 or later, or macOS 11 or later
- **Network:** Internet connectivity required for Streamlit Cloud deployment and GitHub integration

B.2 Software Requirements

The system is built using Python and several supporting libraries for deep learning, signal processing, and deployment.

- **Python:** Version 3.10 or higher
- **PyTorch:** Version 2.0 or higher (CUDA-enabled for training, CPU version for cloud inference)
- **Streamlit:** Version 1.25 or higher for web application deployment
- **WFDB:** Version 4.1 or higher for reading ECG signal files
- **OpenCV & Pillow:** Used for ECG image processing and extraction pipeline
- **NumPy & Pandas:** For numerical computation and data handling
- **Matplotlib & Seaborn:** For visualization of ECG signals and evaluation results
- **ReportLab:** For generating PDF-based diagnostic reports
- **Poppler:** Required for converting PDF ECG reports into images
- **GitHub:** Used for source code management and deployment integration with Streamlit Cloud

C. Generated PDF Report Output

ECG Analysis Report

ECG Abnormality Detection System

Model v1.0 | Analysis time: 2026-03-28 08:25:09

Model Result

Prediction	Abnormal
Classification	**ABNORMAL**
Abnormal Probability	100.0%
Normal Probability	0.0%

Explainability Summary

Visualization Method	12-Lead Global Heatmap + Per-Lead Grad-CAM
Most Activated Lead	V2
Activation Score	0.1084
XAI Method	Grad-CAM + Saliency

The model focused most strongly on lead V2, suggesting pathological patterns in that lead's territory.

Figure 3: Global Heatmap (12-Lead)

Rows = leads, columns = time. Warm colours (red) indicate high Grad-CAM activation; cool colours (blue) indicate low activation.

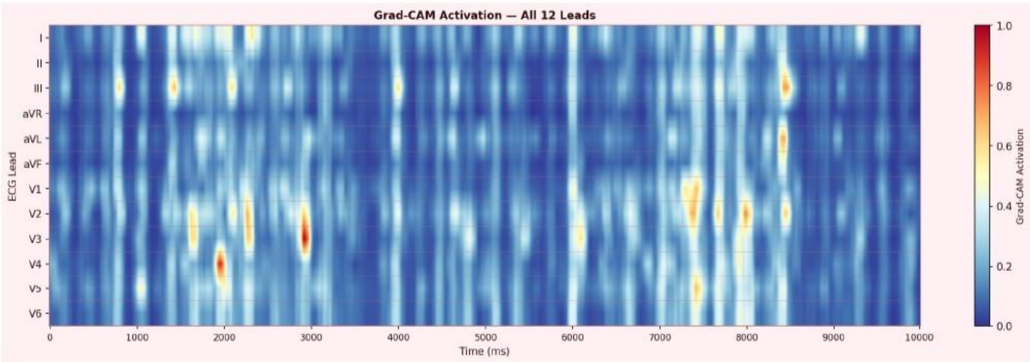


Figure 1: Grad-CAM Activation Map

Colour-coded background segments show where the model focused most strongly. Red = high activation, blue = low activation.

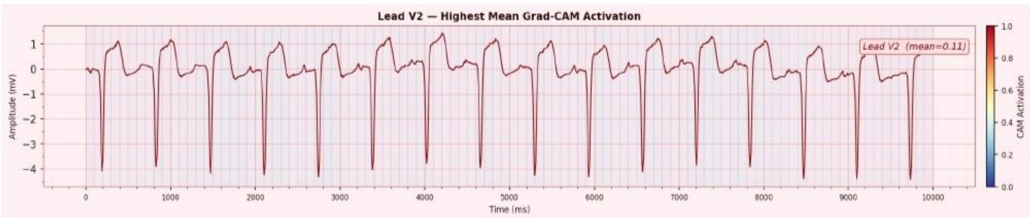
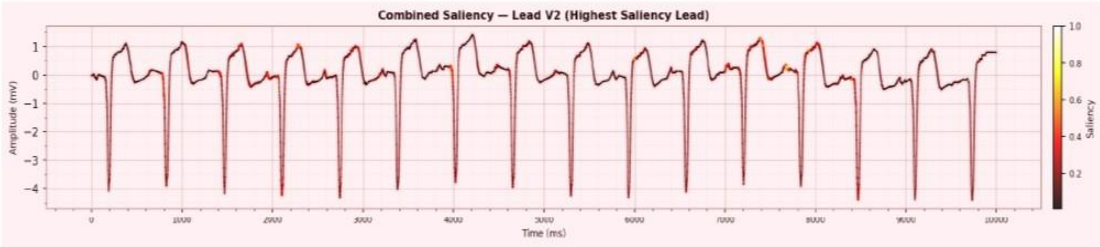


Figure 2: Saliency Map

Individual samples are coloured by gradient magnitude. Hot colours (white/yellow/red) mark the most influential time points.

Fig 10.1: Generated PDF Report Output



Patient Explanation

The AI analysis has detected strong indicators of cardiac abnormality in your ECG. These patterns may be consistent with myocardial infarction (heart attack) or severe ischemic injury. This is a high-priority finding — please seek urgent medical evaluation immediately.

Clinical Summary

AI model output: Abnormal classification. The model's attention (via Grad-CAM) is focused on features consistent with ischemic injury patterns — potential ST-segment deviation, T-wave changes, or morphological QRS abnormalities. Differential includes STEMI, NSTEMI, unstable angina, or non-ischemic cardiomyopathy. Clinical correlation with patient history, symptoms, troponin levels, and repeat ECG is strongly advised. The 12-lead model incorporates spatial information from all lead vectors, providing higher specificity for localizing territory of injury.

Confidence Note

Model confidence: very high (100.0%). Model: 12-lead CNN trained on PTB-XL (500 Hz). This AI prediction is for research and educational use only. A qualified clinician must validate all findings.

Technical Metadata

Signal Length	5000 samples
Duration	10.0 s
Sampling Rate	500 Hz
Leads Analyzed	12
Lead Names	I, II, III, aVR, aVL, aVF, V1, V2, V3, V4, V5, V6
Model Architecture	N/A
Model Version	N/A

Disclaimer

This system is a research and decision-support tool intended for educational and investigational purposes only. It is NOT a medical device and must NOT be used as a sole basis for diagnosis, treatment, or clinical decision-making. The predictions generated by this system are based on patterns learned from historical ECG data and may not generalize to all patient populations, recording devices, or clinical settings. A qualified medical professional must always interpret ECG findings in conjunction with clinical history, physical examination, and additional diagnostic tests. The developers assume no responsibility for clinical outcomes resulting from the use of this system.

Fig 10.1: Generated PDF Report Output

D. QR Code

The QR code below links to the project GitHub repository which contains the source code, trained model weights, dataset access instructions, the final PDF report, the project presentation, and a demonstration video.

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ECG-Based Detection of Myocardial Infarction and Ischemic Abnormalities Using Deep Learning with Explainable AI and Web Deployment

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***Abstract*— Heart diseases are among the most common causes of death in the world. Myocardial infarction (MI) requires fast detection with electrocardiogram (ECG) readings. In this paper, we present an end-to-end deep learning system for automated MI and ischemic abnormality detection based on ECG readings with explainability and web deployment. Three models of convolutional neural networks (CNNs) have been trained with one-dimensional inputs from the PTB-XL dataset (21,799 entries, 500 Hz, 10 seconds): 12-Lead CNN with multi-lead digital ECG input, Lead II CNN with single-lead ECG input, and Image CNN with scanned ECG images. 12-Lead CNN outperformed other models with ROC-AUC = 0.8991, sensitivity = 0.8674, specificity = 0.7580, and F1 score = 0.8436. Explainability with Grad-CAM [16] and saliency maps has been incorporated that conforms with the standard ECG pattern interpretation. All of our models have been integrated in an easily accessible web application with Streamlit Cloud to allow WFDB file, CSV, and image input with automatic route selection and report generation.**

***Keywords*— convolutional neural network, deep learning, electrocardiogram, explainable AI, Grad-CAM, myocardial infarction, PTB-XL, Streamlit, web deployment.**

I. Introduction

According to the WHO, cardiovascular disease is the leading cause of deaths worldwide, responsible for 17.9 million deaths every year [28]. When blood flow to part of the heart muscle becomes blocked, usually by a clot, and cannot be treated quickly, irreversible damage to cardiac tissues can occur. This condition is called myocardial infarction (MI). ECG is the most commonly used diagnostic method for detecting MI, indicating ischemia or infarction with changes like elevated/depressed ST segment, inverted T wave, and abnormal Q waves.

An expert interpretation of ECG is required for reliable diagnosis; however, there may not always be a cardiologist nearby in underserviced areas, such as rural hospitals. Therefore, automated methods with high precision and recall could serve as an invaluable tool in MI detection [18].

There have been numerous previous works in this field that showed strong classification capabilities on benchmark datasets; yet, most of those projects lacked web application interfaces and explainability for clinicians to verify predictions or could not deal with multiple ECG input modalities in one pipeline.

Our contribution lies in: (1) creating three CNN models for each of the input modalities mentioned; (2) applying explainability through Grad-CAM [16] and saliency maps validated by clinical patterns; (3) integrating all models in a publicly accessible web application using Streamlit Cloud; (4) evaluating our models on PTB-XL dataset [14] and proving its feasibility through ablation studies regarding sampling frequency and dataset size.

II. Related Work

Before deep learning became popular, machine learning techniques relied mostly on hand-crafted features. Zhao et al. [2] proposed combining ECG and VCG features for myocardial ischemia detection with a support vector machine (SVM), reaching 90% of accuracy. At the same time, they needed VCG devices which were not readily available to everyone. Sraitih et al. [8] conducted experiments with different classifiers (SVM, k-NN, and Random Forest) under noise conditions for PTB dataset, showing that Random Forest worked best overall, but none of these techniques could be applied to raw waveform learning.

Deep learning has proved to be much more efficient than machine learning for MI detection [21]. Chen et al. [1] employed a 100-layer ResNet CNN on PTB-XL dataset for MI classification and obtained AUC $\approx$ 0.96 without any form of explainability. Hammad et al. [15] also used a ResNet CNN for multi-class classification of cardiac diseases on PTB-XL, obtaining AUC $\approx$ 0.95, yet no explainability has been provided. Hasbullah et al. [4] proposed a hybrid CNN-BiLSTM model with AUC = 0.91 for PTB-XL classification, yet failed to provide explainability. On the other hand, Acharya et al. [20] developed a 9-layer deep CNN to classify five heartbeat categories from ECG signals from MIT-BIH database [26]; they have demonstrated that CNNs can accurately classify arrhythmias from ECG signals. Mousavi and Afghah [19] tackled heartbeat classification problem on MIT-BIH ECGs using sequence-to-sequence deep learning and reached state-of-the-art results both in inter- and intra-patient settings.

In another related work, Wu et al. [3] have reached 0.97+ AUC for ST-elevation MI detection with hospital data, yet did not deploy an explainability module. Moreover, Gustafsson et al. [10] have reached 0.99 C-statistic for STEMI detection on a proprietary ECG dataset from Swedish emergency departments with a 492,000-entry ensemble ResNets model. Gragnaniello et al. [6] performed real-time MI classification on edge hardware while Chen et al. [7] used a multi-domain feature fusion CNN to detect MI and reached 99.9% precision in 2025, although this technique requires complex feature engineering and does not provide a web interface.

Yousuf et al. [9] used 1D Lead II signals converted into 2D Gramian Angular Field images and trained a 2D CNN on MI detection; they reached 99.6% of accuracy in inferior MI detection, although the model cannot be generalized to real ECG scans. Sheth et al. [5] proposed a lightweight CNN-LSTM architecture working on time-frequency representation for web deployment, reaching 83% accuracy. On the other hand, Pawelczyk et al. [12] showed how important it is to use counterfactual explanations to validate model outputs in an ensemble ResNets framework for PTB-XL. Selvaraju et al. [16] have presented an explainable architecture with Grad-CAM which we use in our project. Finally, Kshama et al. [17] provided an ECG-Image-Kit tool that synthesizes ECG images, allowing us to develop the image input pipeline.

To the best of our knowledge, we have never seen any research work tackling all three input modalities together in the literature along with explainability and web application deployment on open-access data.

III. DATASET AND PREPROCESSING

A. PTB-XL Dataset

In this work, we use the PTB-XL [14] database hosted by Physionet [25]. The PTB-XL is comprised of 21,799 to 21,837 clinical records of 12-lead ECG readings from 18,869 to 18,885 patients, each 10 seconds long with a sampling rate of 500 Hz, thus, 5000 samples in total. SCP-ECG superclasses: NORM, MI, STTC, CD, and HYP, have been used to annotate records where MI and STTC are considered positive (indicating MI and ischemia) while the rest are negative. Splitting ratios for the train, validation, and test sets were 70%, 15%, and 15% respectively at the level of patient ids to avoid overfitting.

B. Signal Preprocessing

First of all, ECG signals from digital sources such as Waveform Database Format (WFDB) or CSV files need z-score normalization of each individual lead: $x' = (x - \mu) / (\sigma + 1e\text{-}8)$ where μ and σ stand for the mean and standard deviation of the particular lead signal respectively. Since PTB-XL recordings are already clean clinical ECGs recorded at 500 Hz, no bandpass filtering is required. For image-derived ECG signals, the 4th order Butterworth bandpass filter with 0.5 - 40 Hz cut-off has been applied to remove the effect of the extraction pipeline on ECGs.

C. Synthetic Image Generation

There is no large-scale publicly labelled ECG image dataset suitable for our classification scheme. To generate images suitable for supervised learning, we followed the ECG-Image-Kit procedure [17]. The ECG signal was converted to a fake ECG paper image with pink grid, Gaussian noise, random rotation up to ±1°, and Gaussian blur; then the rhythm strip was extracted from the resulting image with our pipeline. The domain gap between synthesized training ECG images and scanned ECG images used in testing is the main reason why Image CNN performed worse than the rest.

D. Image Extraction Pipeline

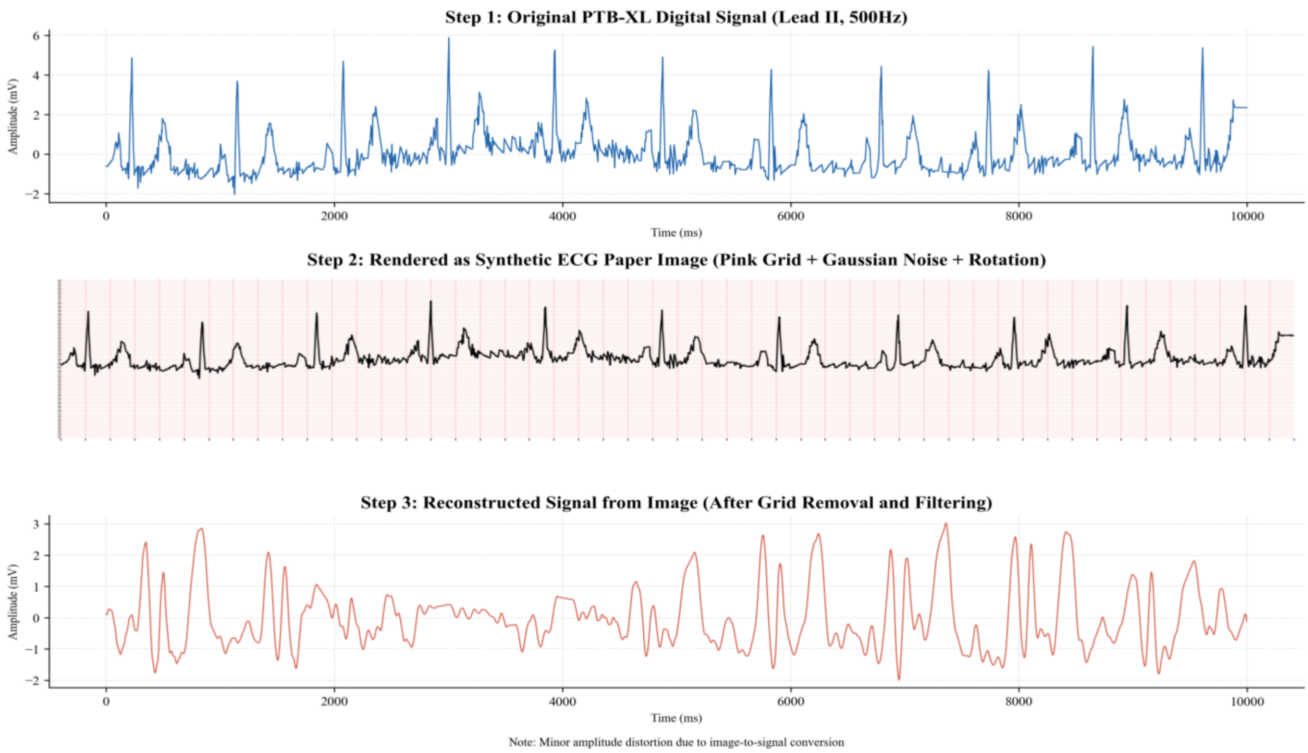

Fig. 1: Synthetic ECG image generation pipeline.

When an image is uploaded, it needs to pass through a series of ten steps before being fed to our models in order to extract the waveform: (1) loading the image or converting the PDF at 200 DPI with the help of Poppler; (2) image orientation from portrait to landscape; (3) converting to grayscale and denoising with Gaussian filter; (4) applying adaptive threshold to make the image binary; (5) using morphological operation with 40×1 kernel to remove horizontal grid lines without altering QRS complexes; (6) extracting rhythm strip from the bottom of the image within 55–76% of rows; (7) using a weighted center of gravity extraction algorithm per image column; (8) applying 4th order Butterworth bandpass filter with 0.5-40 Hz cutoff frequencies; (9) resampling to 5000 samples; (10) z-score normalization. Resulting tensor is (1, 5000) shaped and has float32 precision type.

IV. CNN ARCHITECTURES

Each model implements 1D CNN architecture trained on a fixed sequence length of 5000-time steps (10 s at 500 Hz). 1D CNN has shown state-of-the-art performance in biomedical signal classification [22]. Each convolutional block includes 1D convolution, batch normalization, ReLU activation, max pooling (kernel 2), and dropout (rate 0.3). The classification head is a two-layer fully connected network with one sigmoid activation for binary classification.

A. Lead II CNN

The Lead II CNN takes input of shape (1, 5000). It has three convolutional blocks with 32, 64, and 128 filters with kernel size 7, 5, and 5. After flattening, there is a fully connected (FC) layer of 256 units followed by ReLU activation and dropout. The classification threshold is 0.48 (tuned by maximizing F1 score).

B. 12-Lead CNN

The 12-Lead CNN takes input of shape (12, 5000). It has four convolutional blocks with 32, 64, 128, and 128 filters and kernels of 7, 5, 5, and 3. The added convolutional block allows learning of complex interactions between leads from the richer multi-lead representation. The classification head is the same as in Lead II CNN. Threshold: 0.43.

C. Image CNN

The Image CNN uses the architecture of the Lead II CNN, receiving input of shape (1, 5000) via the image pipeline. Separate model trained on synthetic ECG images. Threshold: 0.91. This relatively high value is caused by overconfidence on the synthetic training data (known calibration issue in such setting).

V. MODEL TRAINING

Models were implemented using PyTorch [29] framework and trained on NVIDIA CUDA GPU. Adam optimizer [23] was used with weight decay of 1e-4 and learning rate of 1e-3. The ReduceLROnPlateau learning rate scheduler was configured to half the learning rate when validation loss stops improving for five consecutive epochs. Training was performed for 50 epochs with the batch size of 64, using binary cross-entropy with logits as the loss function. The classification thresholds were tuned based on F1 score optimization.

An ablation study was conducted in order to validate three key design choices of the 12-Lead CNN model. First, increasing sampling frequency from 100 Hz to 500 Hz led to improvement of 0.038 AUC (0.8991 vs. 0.8612), indicating that a higher temporal resolution helps capture relevant ECG information. Second, decreasing the number of data samples to 10% decreased AUC by 0.078 (0.8213), proving the importance of the full-size PTB-XL dataset. Third, GPU training led to a 2-fold reduction in epoch time (from ~20 min to <2 min), while producing the same model quality, compared to CPU training.

VI. SYSTEM ARCHITECTURE AND DEPLOYMENT

The proposed system implements a single-tier architecture, whereby all of its components operate within a single Streamlit application: loading of an ECG, input routing, preprocessing, CNN inference, Grad-CAM generation, saliency calculation, natural language interpretation, and generating of the final PDF report. There is no separate backend server, as this architecture was forced due to Streamlit Cloud restriction to only one application.

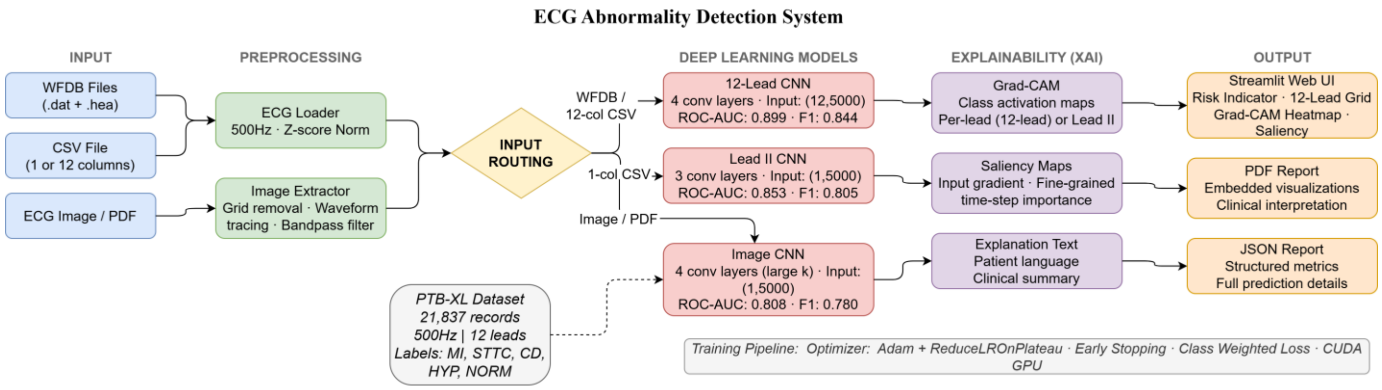

Fig 2: System Architecture

At the beginning, the system was designed with a backend server, implementing model inference using FastAPI framework. The deployment stage involved changing the architecture to the proposed one in order to simplify operations and eliminate network latency. Model inference is done on CPU within a cloud environment and takes ~15-25 seconds, including Grad-CAM calculation and PDF generation. Model weights (ecg_cnn_1lead.pth: 82 MB; ecg_cnn_12lead.pth, ecg_cnn_image.pth: 41 MB each) are stored on Hugging Face Hub [30] and automatically downloaded during the first use of the model per session (takes ~30-60 seconds).

The application is deployed from https://github.com/Abdullah1607/ECG-Abnormality-Detection-System and is live at https://ecg-insight.streamlit.app. Table I summarizes the input routing logic.

TABLE I

INPUT ROUTING LOGIC

Input Format	Model	AUC	Status
WFDB (.dat+.hea)	12-Lead CNN	0.899	Primary
CSV (12 columns)	12-Lead CNN	0.899	Primary
CSV (1 column)	Lead II CNN	0.853	Primary
Image / PDF	Image CNN	0.808	Experimental

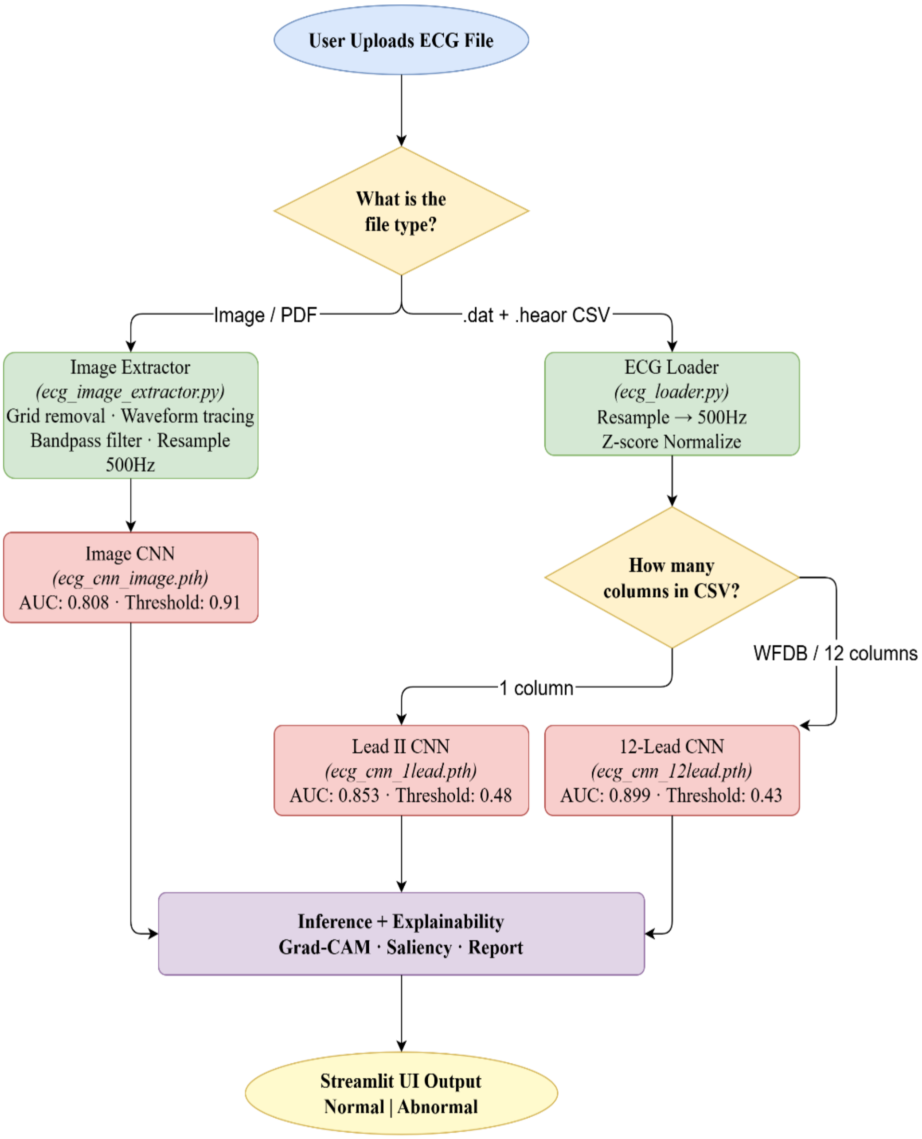

Fig 3: Input Routing Logic

VII. Explainability Module

Explaining the model predictions is done via Grad-CAM [16] and input gradient saliency maps. Grad-CAM, built upon class activation maps (CAM) ideas [24], calculates the weighted sum of feature maps from the last convolutional layer using the gradients of the class logit with respect to these maps as weights. Activation map is upscaled to 5000 samples using linear interpolation and then normalized to [0, 1]. Saliency maps are calculated via dL/dx, i.e., gradient of the class logit with respect to the input vector. All of the hooks are cleared after the calculations to free up memory space.

For 12-Lead CNN, four types of explainability outputs are generated: (1) overall heatmap of the Grad-CAM activation values in all leads and across all time steps; (2) identification of the lead with the highest activation; (3) visualization of a 12-lead ECG grid, where each lead has color overlays corresponding to the Grad-CAM values; (4) a combined saliency map averaged over leads and displayed on the lead with the highest average activation. For Lead II CNN and Image CNN, there are three types of explainability outputs: Grad-CAM heatmap applied to the original ECG and visualized with the RdYlBu_r colormap; saliency map presented as a scatter plot with hot color scale; original ECG annotated with a prediction label and confidence score.

Grad-CAM heatmaps of the 12-Lead CNN consistently show activation peaks around the ST-segment and T-waves in leads II, V4, V5, and V6, which is consistent with the location of abnormalities observed in positive cases of MI. For cases of inferior MI, the highest activation was in leads II, III, and aVF, which is consistent with the established clinical pattern for MI resulting from right coronary artery occlusion. Saliency maps show peak activation around the QRS complex onset and the beginning of the ST segment. The alignment between the Grad-CAM findings and known clinical ECG markers proves that the model learns clinically relevant features rather than dataset artifacts.

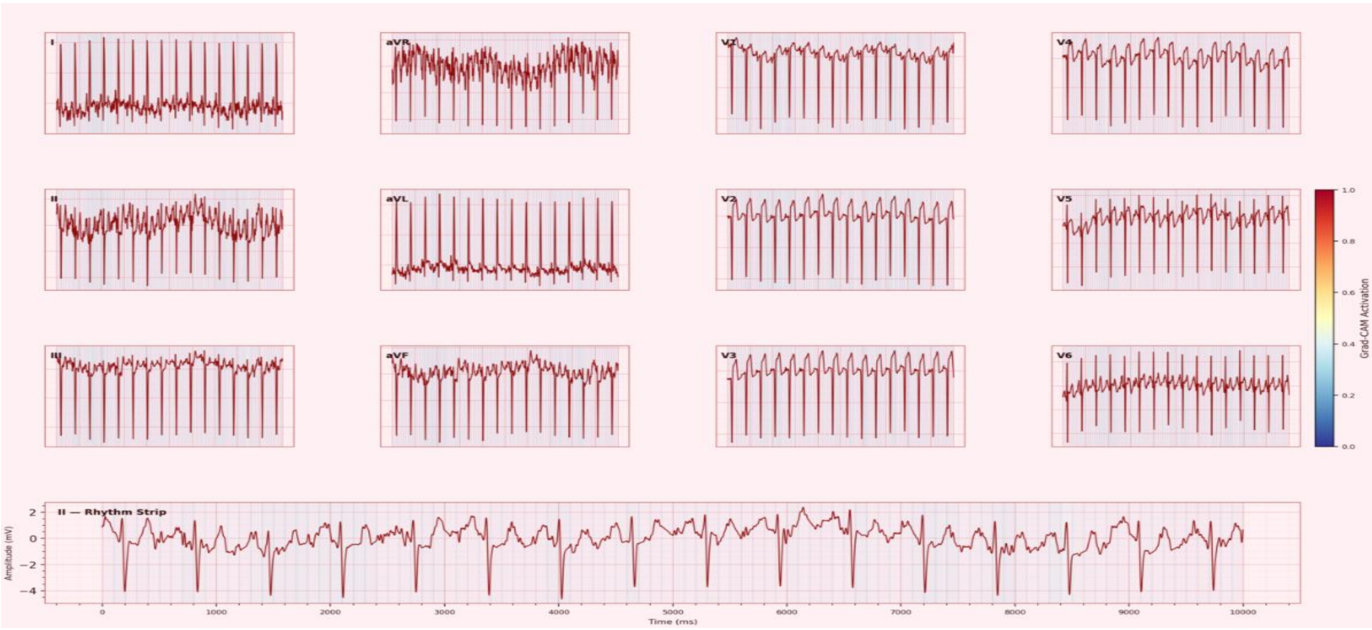

Fig 4: 12-Lead ECG – Grad Cam Overlay

VIII. Results

Evaluation results for three models are presented below. Each model was tested on the test dataset of ~3,275 records. The thresholds were selected based on F1 score maximization on the validation set. ROC-AUC, Sensitivity, Specificity, and F1 score were calculated using scikit-learn [27].

TABLE II

Model Performance on PTB-XL Test Set (~3,275 Records)

Metric	Lead II CNN	12-Lead CNN	Image CNN	Threshold
ROC-AUC	0.8528	0.8991	0.8083	—
Accuracy	0.7682	0.8099	~0.75	—
Sensitivity	0.8500	0.8674	0.8260	—
Specificity	0.6634	0.7580	0.6180	—
F1 Score	0.8045	0.8436	0.7800	—
Threshold	0.48	0.43	0.91	—

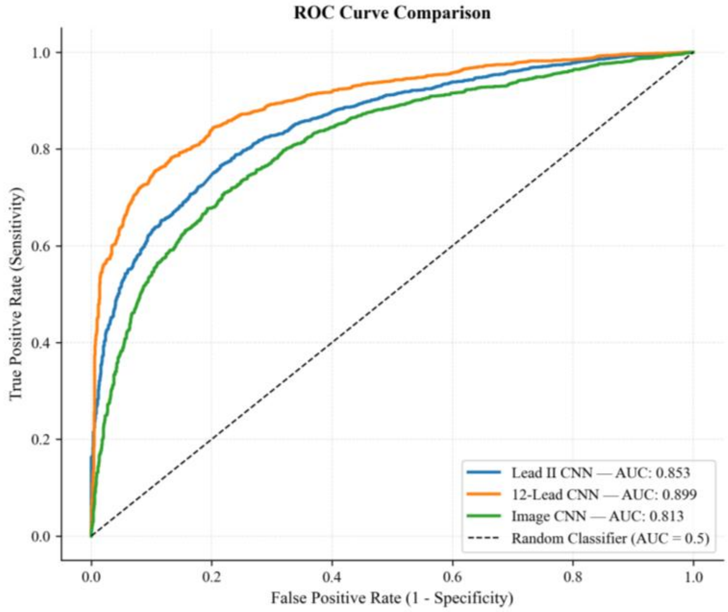

Fig 5: ROC Curve Comparison

The 12-Lead CNN model achieved the best results across all five metrics, demonstrating 0.047 difference in AUC and 0.095 difference in specificity. High specificity means that the model minimizes unnecessary investigations triggered by the false alarms. Both digital models were tuned for maximizing sensitivity since the detection of a MI is more costly than a false alarm. The Image CNN had a much higher threshold (0.91) because of the calibration issues due to distribution shift between synthetic and real-world datasets.

TABLE III

ABLATION STUDY RESULTS (12-LEAD CNN)

Configuration	AUC	F1	Observation
500 Hz, full dataset, GPU	0.8991	0.8436	Final configuration
100 Hz, full dataset, GPU	0.8612	0.8101	−0.038 AUC
500 Hz, 10% dataset, GPU	0.8213	0.7734	−0.078 AUC
500 Hz, full dataset, CPU	0.8991	0.8436	10× slower train

IX. WEB APPLICATION

The web application, deployed at https://ecg-insight.streamlit.app, adopts a pink medical theme and includes the visualization of ECG paper grids. A clinical risk indicator based on the predicted probability color codes the output into green (< 0.3 – low risk), orange (0.3-0.6 – moderate risk), and red (> 0.6 – high risk) categories. The probability bars provide precise probability scores per each class.

For each result, two tabs are automatically generated. The patient version tab presents the result using layman terms without any clinical terminology. The clinical version includes model-specific jargon along with information about model performance and the limitations of AI-based results. In case the input is an ECG image, an additional research tab is generated explaining the synthetic training strategy and marking the result as experimental. Finally, a PDF report summarizing all results, ECG plot, Grad-CAM heatmap, and saliency maps may be downloaded.

X. TESTING AND VALIDATION

Testing was done at three stages. Unit testing confirmed functionality of 14 independent modules, including z-score normalization, WFDB and CSV data import, lead routing, Lead II CNN inference, 12-Lead CNN inference, image extraction pipeline output size, Image CNN inference, Grad-CAM output shape (12, 5000), saliency map generation, and PDF validation. All 14-unit tests passed successfully.

Integration tests validated nine separate workflows: importing ECG via WFDB/CVS to produce classification result, Grad-CAM, and PDF export; fully visualizing a 12-column CSV; performing Lead II classification on a single column CSV; uploading an image, performing extraction, and producing Image CNN result with a research note; downloading a PDF report with embedded visualizations; displaying a clinical risk indicator for the highest-probability MI input; generating patient and clinical interpretation tabs; and automatic Streamlit Cloud rebuild upon GitHub push. All 9 integration tests succeeded.

System tests confirmed robustness of the codebase in 6 separate use cases: graceful handling of unsupported file types; graceful return in case of corrupt CSV data; handling low-quality ECG image and performing the full pipeline while noting it as experimental; accurate application of thresholds for all three classification problems (0.48, 0.43, 0.91); performing CPU-based inference on the Streamlit Cloud and returning results in 15-25 seconds; and acceptance testing by non-technical users to confirm proper functioning of all features, from uploading an image to the PDF report download. All 6 system tests passed successfully.

XI. Discussion

The ROC-AUC 0.8991 and F1 0.8436 of the 12-Lead CNN results are competitive when compared with other CNN-based systems on the PTB-XL database that lack explainability and deployment. The work of Chen et al. [1] achieves AUC 0.96 for classification with a ResNet-based architecture, suggesting potential improvements with more sophisticated designs, but their system lacks any explainability and web deployment. The 12-Lead CNN's lower AUC reflects the trade-off of designing an explainable and deployable 1D CNN system, as opposed to a deep model.

TABLE IV

Comparison With Published Systems on PTB-XL

System	Architecture	AUC	Deploy	Explainability
Chen et al. [1]	ResNet	0.960	No	None
Hammad et al. [15]	Deep CNN	0.950	No	None
Hasbullah et al. [4]	CNN-BiLSTM	0.910	No	None
Pawelczyk et al. [12]	Custom	N/A	No	Counterfactual
Proposed (12-Lead)	1D CNN	0.899	Yes	Grad-CAM + Saliency
Proposed (Lead II)	1D CNN	0.853	Yes	Grad-CAM + Saliency
Proposed (Image)	1D CNN	0.808	Yes	Grad-CAM + Saliency

Comparing our results to those reported in previous literature, it is clear that the 12-Lead CNN AUC of 0.8991 lags behind some published works. However, it stands out due to the web deployment capability and Grad-CAM explainability, which no other system in the table provides. The reasons for this disparity are the reliance on a simpler 1D CNN architecture, using a public PTB-XL dataset with less clean labeling, and including STTC records to the target classes. Published systems achieve their high performance through more complex architectures, using internal/private ECG data, and excluding the entire STTC class.

The ROC-AUC 0.8528 of the Lead II CNN provides a valuable option for screening ECGs with only single-lead recordings. As mentioned earlier, the specificity of 0.663 represents its weakness. However, it is appropriate for an ECG screening application, since false positives will trigger further clinical evaluation, not direct treatment.

Aligning the Grad-CAM activation map with the known ECG markers for specific MI is important confirmation that the model has learned the necessary features. It activates the ST-segment and T-wave regions of the anterior leads (V1-V3) to detect anterior MI, and leads II, III, and aVF to detect inferior MI. Therefore, the model did not exploit dataset biases.

The key drawback of the proposed framework is the dependency of the image pathway on the synthetic training pipeline. The variation in actual clinical ECG printouts is much greater, as hospitals use various printing equipment and practices, leading to differences in print qualities, color tones, and grid resolutions. Moreover, the low threshold (0.91) and low specificity (0.618) for the image pathway make it unsuitable for clinical deployment without further work.

The second problem relates to the PTB-XL dataset used for training. Despite the relatively large size and the existence of multiple classes, its records come from a single European hospital. Consequently, PTB-XL does not capture the global population's variability. Furthermore, the dataset contains records labeled by the authors themselves as borderline, i.e., annotated at very low confidence. It contributes to the low specificity of the Lead II CNN, since negative samples include mislabeled MI instances.

Comparing with other literature works is difficult because many researchers use different datasets, define different targets (e.g., STEMI-only), and apply different thresholds when calculating AUC. On top of that, many works report AUC > 0.95 using train/overlap test sets. In contrast, our evaluation used a stricter splitting, considering the entire class of MI+STTC, with a non-overlapping set of patients. Considering the challenge posed to the model by the task and the dataset, we consider the results of the Lead II (ROC-AUC = 0.8528, F1 = 0.7883) and 12-Lead (ROC-AUC = 0.8991, F1 = 0.8436) CNNS competitive.

XII. Conclusion

In this paper, we proposed a fully functional web platform for myocardial infarction and ischemic abnormality detection from ECG data. Three distinct models, trained on the publicly available PTB-XL database, handle multi-lead, single-lead, and ECG image inputs in a uniform pipeline. The 12-Lead CNN obtained ROC-AUC 0.8991 and F1 0.8436, comparable with other published works that lack both explainability and web-deployability. Additionally, the Grad-CAM activations demonstrated agreement with the known ECG markers of MI. Finally, the proposed platform is publicly available as a single-tier Streamlit Cloud web app.

Future work needs to collect real-life ECG image datasets to bridge the domain gap in the Image CNN. Extending classification to arrhythmia and conduction disorders increases the applicability of the models, as deep neural networks proved themselves on a variety of physiological signals [21]. Transformer-based architectures with inbuilt attention mechanism provide more intuitive explanations compared to post-hoc Grad-CAM analysis. Evaluating on Indian ECGs will confirm generalization to local population demographics and ECG recording practices.

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QUALITY ANALYSIS AND ALIGNMENT OF STUDENT PROJECTS WITH PROGRAM OUTCOMES (POs) AND SUSTAINABLE DEVELOPMENT GOALS (SDGs)			
Course Name:	PROJECT WORK – II	**AY**	2025-26
Course Code:	3PW868CS	**Semester**	VIII
Name of the Guide:	Ms. Sana Mateen	**Class**	CSE - A

PROJECT TITLE
ECG-Based Detection of Myocardial Infarction and Ischemic Abnormalities Using Deep Learning and Web Deployment

STUDENTS

S. No	HT No	Name of the Student
1.	160722733016	Mohammed Faisal Mohiuddin
2.	160722733029	Mohammed Hussain Moheet
3.	160722733056	Mohammed Abdullah

Course Outcome: After the successful completion of this course, the student will be able to:

CO No	Course Outcome	BTL	POs, PSOs Mapped
3PW868CS.1	Demonstrate the ability to synthesize and apply the knowledge and skills acquired in the academic program to the real-world problems.	**Apply**	PO1, PO2, PO3, PO6, PO8, PO11, PSO1, PSO2, PSO3
3PW868CS.2	Analyze and formulate the problem, and design solution of the selected project.	**Analyze**	PO1, PO2, PO3, PO4, PO5, PO6, PO8, PO11, PSO1, PSO2, PSO3
3PW868CS.3	Evaluate different solutions based on economic and technical feasibility.	**Evaluate**	PO3, PO4, PO5, PO6, PO7, PO8, PO11, PSO1, PSO2, PSO3
3PW868CS.4	Effectively plan, develop and deploy a project and confidently perform all aspects of project management.	**Create**	PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3
3PW868CS.5	Demonstrate effective written and oral technical communication skills.	**Create**	PO2, PO5, PO7, PO9, PO10, PO12, PSO1, PSO2, PSO3

CO-PO-PSO MAPPING

3 – High, 2 – Moderate, 1 – Slight

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
3PW868CS.1	3	2	2	-	-	3	-	2	-	-	2	-	3	3	3
3PW868CS.2	3	3	2	3	2	2	-	2	-	-	2	-	3	3	3
3PW868CS.3	-	-	3	3	2	2	2	2	-	-	3	-	3	2	2
3PW868CS.4	-	-	3	3	3	2	2	3	3	3	3	2	3	3	3
3PW868CS.5	-	2	-	-	2	-	2	2	3	3	2	2	2	2	2

CO-PO-PSO MAPPING (With Performance Indicators)

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS0 1	PSO 2	PSO 3
3PW868CS.1	1.1.2 1.2.1 1.3.1 1.4.1	2.1.2 2.1.3 2.2.3 2.3.1 2.4.1	3.1.1 3.1.3 3.1.4 3.1.5 3.2.1 3.2.3 3.3.1 3.3.2	-	-	6.1.1 6.2.1 6.3.1 6.3.2 6.4.1	-	8.2.1 8.2.2 8.2.3	-	-	11.1.2 11.2.1 11.2.2 11.3.1 11.3.2	-	**High**	**High**	**High**
3PW868CS.2	1.1.2 1.2.1 1.3.1 1.4.1	2.1.1 2.1.2 2.1.3 2.2.3 2.3.1 2.4.1 2.4.3 2.4.4	3.1.1 3.1.3 3.1.4 3.1.5 3.1.6 3.2.1 3.2.3 3.3.1 3.3.2 3.4.1 3.4.2	4.1.1 4.1.2 4.1.3 4.2.1 4.3.1 4.3.2 4.3.3 4.3.4	-	6.1.1 6.3.1 6.3.2 6.4.2	-	8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1	-	-	11.1.2 11.2.1 11.2.2 11.3.2	-	**High**	**High**	**High**
3PW868CS.3	-	2.2.2 2.2.3 2.2.4 2.3.2 2.4.2 2.4.3 2.4.4	3.1.5 3.2.1 3.2.2 3.2.3 3.3.1 3.3.2 3.4.1 3.4.2	4.1.2 4.2.1 4.2.2 4.3.1 4.3.2 4.3.3 4.3.4	5.2.1 5.2.2 5.3.1 5.3.2	6.1.1 6.2.1 6.3.1 6.3.2 6.4.2	7.1.1 7.1.2 7.2.1 7.2.2	8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1	-	10.1.1 10.1.2 10.2.1	11.1.2 11.2.1 11.2.2 11.3.2	-	**High**	**Mod**	**Mod**
3PW868CS.4	-	2.3.1 2.3.2 2.4.1 2.4.2 2.4.3 2.4.4	3.1.6 3.2.1 3.2.2 3.3.2 3.4.1 3.4.2	4.1.2 4.1.3 4.2.1 4.2.2 4.3.1 4.3.3	5.1.1 5.1.2 5.2.1 5.2.2 5.3.1 5.3.2	6.1.1 6.2.1 6.3.1 6.3.2 6.4.2	7.1.1 7.1.2 7.2.1 7.2.2	8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1	9.1.1 9.1.2 9.1.3 9.2.1 9.2.2 9.3.1 9.3.2	10.1.1 10.1.2 10.2.1 10.3.1 10.3.2	11.1.1 11.1.2 11.2.1 11.2.2 11.3.2	12.1.1 12.1.2 12.2.1 12.2.2 12.3.1 12.3.2	**High**	**High**	**High**
3PW868CS.5	-	2.2.1 2.2.2 2.2.3 2.2.4	-	-	5.1.1 5.1.2 5.2.1 5.2.2 5.3.1 5.3.2	-	7.1.1 7.1.2 7.2.1 7.2.2	8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1	9.1.1 9.1.2 9.1.3 9.2.1 9.2.2 9.3.1 9.3.2	10.1.1 10.1.2 10.2.1 10.3.1 10.3.2	11.1.1 11.1.2 11.2.1 11.2.2 11.3.2	12.1.1 12.1.2 12.2.1 12.2.2 12.3.1 12.3.2	**Mod**	**Mod**	**Mod**

Project Specific Learning Outcomes

LO#	Outcome Description
1	Design and implement CNN-based models for detection of myocardial infarction and ischemic abnormalities from ECG data.
2	Process and analyze multi-format ECG inputs (WFDB, CSV, and images) for robust model development.
3	Evaluate model performance using metrics such as ROC-AUC, sensitivity, specificity, and F1-score.
4	Apply explainable AI techniques (Grad-CAM and saliency maps) to interpret model predictions.
5	Develop and deploy a web-based system for automated ECG analysis and clinical report generation.

PO -WK Mapping:

PO#	PO Description	Mapped WKs	PO#		Mapped WKs
PO1	Engineering Knowledge	WK1, WK2, WK3, WK4	PO7	Environment and sustainability	WK5, WK7
PO2	Problem Analysis	WK1, WK2, WK3, WK4	PO8	Ethics	WK7, WK9
PO3	Design / Development of Solutions	WK1, WK2, WK3, WK4, WK5, WK6, WK7	PO9	Individual & Team Work	WK9
PO4	Conduct Investigations of complex problems	WK2, WK3, WK4, WK8	PO10	Communication	WK7, WK9
PO5	Engineering Tool usage	WK2, WK6	PO11	Project Management & Finance	WK5, WK6, WK7
PO6	The Engineer and society	WK1, WK5, WK7	PO12	Life-long Learning	WK8

SDG Mapping:

SDG	Mapped Indicator	SDG	Mapped Indicator	SDG	Mapped Indicator
1 NO POVERTY		7 AFFORDABLE AND CLEAN ENERGY		13 CLIMATE ACTION	
2 ZERO HUNGER		8 DECENT WORK AND ECONOMIC GROWTH		14 LIFE BELOW WATER	
3 GOOD HEALTH AND WELL-BEING	3.4.1, 3.8.1, 3.d.1	9 INDUSTRY, INNOVATION AND INFRASTRUCTURE	9.5.1, 9.c.1	15 LIFE ON LAND	
4 QUALITY EDUCATION		10 REDUCED INEQUALITIES		16 PEACE, JUSTICE AND STRONG INSTITUTIONS	
5 GENDER EQUALITY		11 SUSTAINABLE CITIES AND COMMUNITIES		17 PARTNERSHIPS FOR THE GOALS	
6 CLEAN WATER AND SANITATION		12 RESPONSIBLE CONSUMPTION AND PRODUCTION			

QUALITY ANALYSIS

Project collaboration (Tick ✓ Whichever is applicable and present the details below):

Industry	Research Lab	Professional Society	Community	Alumni	Regulating agency	Govt.	Other
			✓				
Details: Provides a potential AI-based diagnostic support system for early detection of cardiac abnormalities in resource-limited settings.							

Project Achievements:

Does the Abstract (prepared/ reviewed / Approved) have the following stated outcomes mentioned?	Yes

S. No	Description	Details
1.	Prototype development	--
2.	Technology based solution developed	ECG-Based Detection of Myocardial Infarction and Ischemic Abnormalities Using Deep Learning and Web Deployment
3.	Publication	Published in IJRASET, Volume 14, Issue IV (April 2026), Paper ID: IJRASET79667.
4.	Start-up incubation	--
5.	Competition prizes	--
6.	Community / Industry / HEI service	Potential healthcare support system for early diagnosis in clinical and remote settings
7.	SDG Indicators	3.4.1, 3.8.1, 3.d.1, 9.5.1, 9.c.1
8.	Grants / Financial Aid	--
9.	Multi-disciplinary	Integration of Healthcare, Artificial Intelligence, and Web Technologies
10.	Internship / Employment	--
11.	Self-employment skill	--
12.	Use of real-time data	Utilized PTB-XL ECG Dataset, a real-world clinical dataset for training and evaluating the model

Signature of the Guide **Signature of the Project Coordinator**